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1 Introduction

Transplant Network

Despite the continuing and dramatic advances in medicine and health technology, the demand for organs for transplantation drastically exceeds the number of donors. To help this situation, US Congress passed the National Organ Transplant Act in 1984, establishing the Organ Procurement and Transplantation Network (OPTN) to match organ donors to patients with organ needs. Even with all this organizational technology and service in place, there are nearly 94,000 transplant candidates in the US waiting for an organ transplant and this number is predicted to exceed 100,000 very soon.

Focusing on Kidney Exchange

In the US, about 68,000 patients are waiting for a kidney from a deceased donor, the median wait for a matching kidney is three years—unfortunately, some needy patients do not survive long enough to receive a kidney.

Ethical Concerns

Transplantation is a controversial issue with both technical and political issues that involve balancing what is best for society with what is best for the individual. Criteria have been developed very carefully to try to ensure that people on the waiting list are treated fairly, and several of the policies try to address the ethical concerns of who should go on to the list or who should come off.

Political issues

Regionalization of the transplant system has produced political ramifications . Doctors living in small communities, who want to do a good job in transplants, need continuing experience by doing a minimum number of transplants per year. However, the kidneys from these small communities frequently go to the hospitals in the big city.

Task 1: Build a mathematical model for the generic US transplant network(s). This model must be able to give insight into the following: Where are the potential bottlenecks for efficient organ matching? If more resources were available for improving the efficiency of the donor-matching process, where and how could they be used? Would this network function better if it was divided into smaller networks (for instance at the state level)? And finally, can you make the system more effective by saving and prolonging more lives? If so, suggest policy changes and modify your model to reflect these improvements.

Task 2: Investigate the transplantation policies used in a country other than the US. By modifying your model from Task 1, determine if the US policy be would improved by implementing the procedures used in the other country. As members of an expert analysis team (knowledge of public health issues and network science) hired by Congress to perform a study of these questions, write a one-page report to Congress addressing the questions and issues of Task 1 and the information and possible improvements you have discovered from your research of the different country's policies. Be sure to reference how you used your models from Task 1 to help address the issues.

Task 3: Devise a procedure to maximize the number and quality of exchanges, taking into account the medical and psychological dynamics of the situation. Justify in what way your procedure achieves a maximum. Estimate how many more annual transplants your procedure will generate, and the resulting effect on the waiting list.

Task 4: Devise a strategy for a patient to decide whether to take an offered kidney, or to even participate in a kidney exchange. Consider the risks, alternatives, and probabilities in your analysis.

Task 5: Based on your analysis, do you recommend any changes to these criteria and policies? Discuss the ethical dimensions of your recommended exchange procedure and your recommended patient strategy (Tasks 3 and 4). Rank order the criteria you would use for priority and placement, as above, with rationale as to why you placed each where you did. Would you consider allowing people to sell organs for transplantation? Write a onepage paper to the Director of the US Health Resources and Services Administration with your recommendations.

Task 6: From the potential donor's perspective, the risks in volunteering involve assessing the probability of success for the recipient, the probability of survival of the donor, the probability of future health problems for the donor, the probability of future health risks (such as failure of the one remaining kidney), and the postoperative pain and recovery. How do these risks and others affect the decision of the donor? How do perceived risks and personal issues (phobias, irrational fears, misinformation, previous experiences with surgery, level of altruism, and level of trust) influence the decision to donate? If entering a list paired network rather than a direct transplant to the relative or friend, does the size n of the n -paired network have any effect on the decision of the potential donor? Can your models be modified to reflect and analyze any of these issues? Finally, suggest ways to develop and recruit more altruistic donors.

2 Assumptions

- The data and material collected from the internet are reliable.
- The graft survival rate increases as the tissue type mismatch decreases.
- There is no constraint on the number of transplants that can be simultaneously carried out.
- Indirect exchanges are feasible.

3 Symbols used in each task

Task 1

m :	Total number of donors
n :	Total number of recipients
x_i :	donor i
y_j :	recipient j
d_{ij} :	The real distance from x_i to y_j
c_j :	The condition of recipient j
w_{ij} :	Matching degree of x_i and y_j
a_i :	donation center i
b_j :	transplant center j
s_{ij} :	The number of organ transport from a_i to b_j
T :	Average turning round time

Task 2

p_i :	The lowest price that donor i charge from the recipient
q_i :	The highest price that recipient i can afford
a_i :	Pricing tolerance
a_{max} :	Maximum price tolerance
C :	Publicize vigor

Task 3

(k_i, t_i) :	A donor – patient pair.
K_i :	Living donor kidneys compatible with patient t_i .
ω :	Priority in the waitlist in exchange for a live kidney.
P_i :	Strict preferences over $K_i \cup \{k_i, \omega\}$.

Task 4

t_i :	Time interval i
λ_i :	The pain of the patient at time t_i
ξ_i :	The condition of the patient at time t_i
β_i :	The risk of the patient after transplant at time t_i
γ_i :	The number of available organ for patient at time t_i
p_i :	The probability of a better matching kidney occur at time t_i

Task 6

X_i : Factor i that affect the decision of donor

P : The value of decision whether to donate or not

b_i : The weight coefficient of factor i that affect the donor's decision

C : Paired comparison matrix

W : Weight vector

$\bar{W}$: Weight vector after normalized

4 Task 1

4.1 Analysis

Firstly, it is needed to analyze and manipulate the related data to build a model to the OPTN network. So in the following part we analyze the national data of deceased and living donors, national data of deceased living transplants, deceased and living donors by region and deceased and living transplants by region in U.S. by some figures gained from those data.

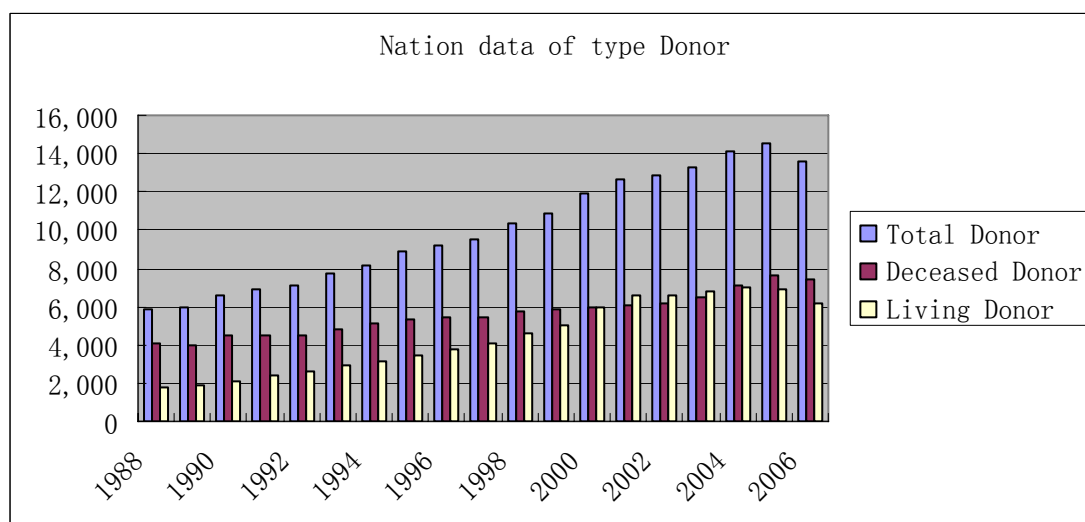


Fig. 1: national data of Deceased and living donors

According to Fig.1, the number of organ donors increased gradually from 5901 in 1988 to 14494 in 2005, while in 2006 it decline to 13577. The figure suggest that people's sense of donation is increased in the U.S. public. However, it is also suggested in the figure that the number of deceased donar didn't have great increase, most of the increase of donoation is attributed to living donor. As deceased donor is a significant part of organ donation, the organ transplant policy in U.S. have its disadvantage in attract deceased donor.

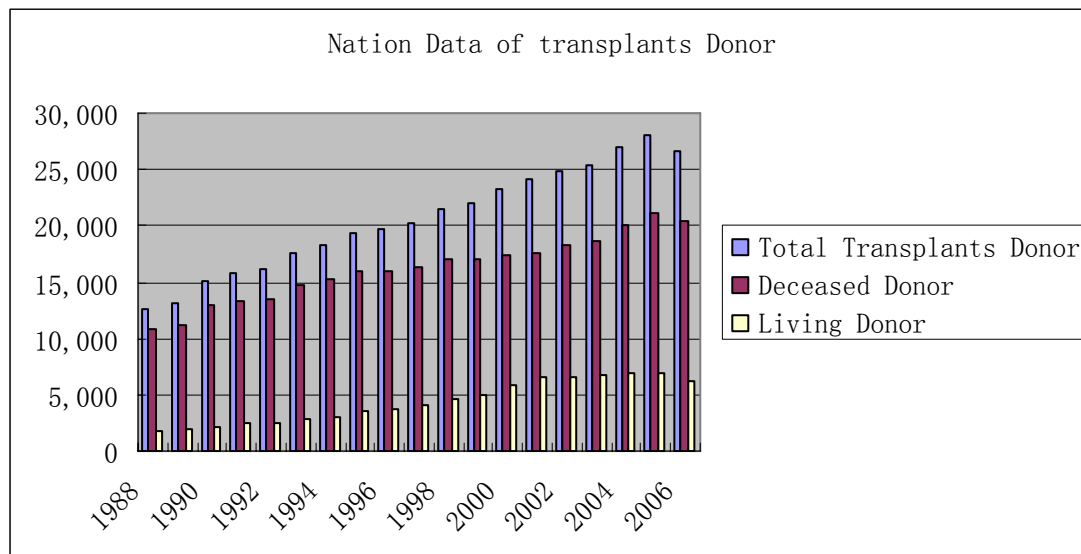


Fig. 2: national data of Deceased and living transplants

According to Fig. 2, the number of organ transplants increased gradually from 12618 in 1988 to 28112 in 2005, while in 2006 it decline to 26689, in which the great mass of transplants are deceased donar transplants. The number of deceased donor make up of about 80% of the total number of transplants. So the deceased organ transplant is a crucial part of argan transplants.

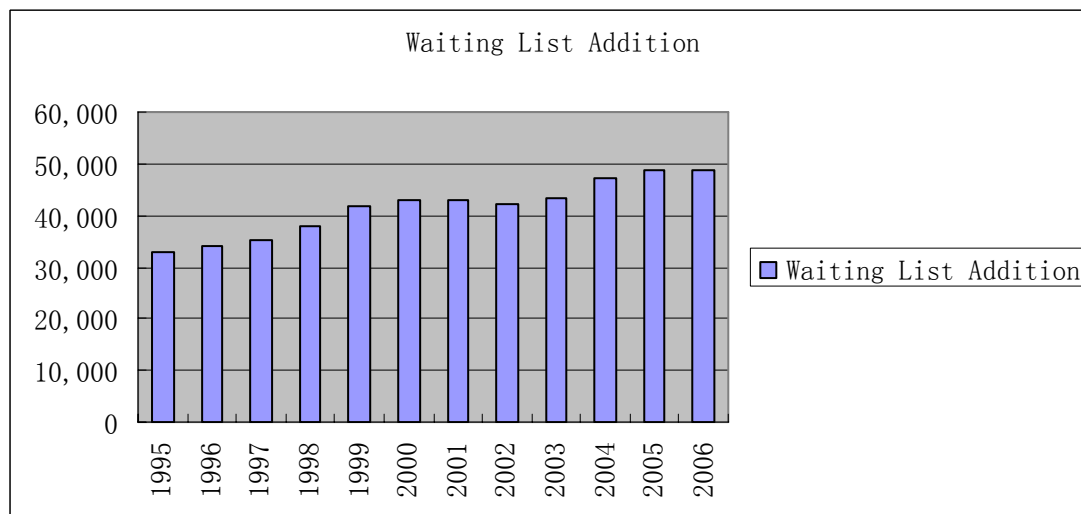


Fig. 3: Waiting List Addition

According to Fig. 3, the number of patients waiting for the organ transplant is more and more, it cause the waiting list grow longer and longer. The potential bottlenecks for efficient organ matching are the following three factors:

- Shortage of donar

Due to the ethical and moral constraints, people are not positive to donate their organ. And some people have some superstition thought such as keep intact remains after death.

- Technological restrictions

Currently, the transplant technology is not so good and need some development. For example, if the technology can change the mismatched organ into matched organ by change some features of the organ, then the match probability can increased sharply and enhance the success rate of the transplants.

- The matching is unefficient

As the restrictions of organ transplant is strict, and considering the exclusive reaction and incompatible phenomenon after organ transplant, it is very difficult to find a suitable organ to transplant.

Model 1 Maximum matching model (MM)

Consider there are A donors and B recipients in a range(nation,region or state), where $B > A$. x_i is donor i and y_j is recipient j in waiting list. The following figure is about the matching of supply and demands.

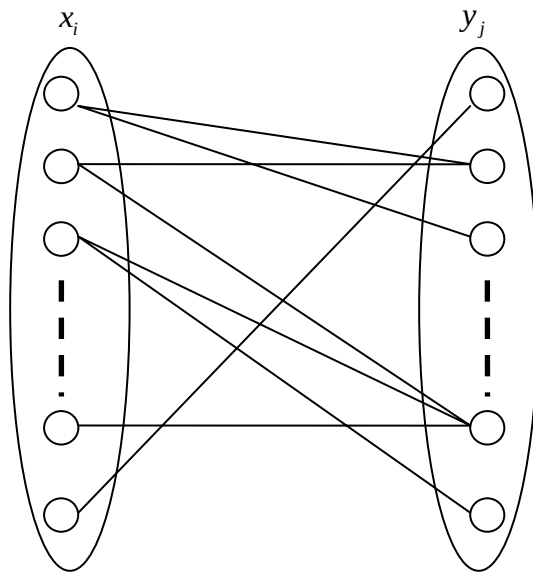


Fig. 4: figure of matching of supply and demand

Build the optimizing model according to Fig. 4.

$$\text{Set } t_{ij} = \begin{cases} 1, & x_i \text{ is matched with } y_j \\ 0, & \text{otherwise} \end{cases}$$

Optimal objective:

Considered the matching level, emergency of transplants of patients and the distance from donor to recipient to maximize the number of matching with A donors.

d_{ij} : the distance between donor i and recipient j

c_j : the emergency of recipient j

Where $c_j \in [0,1]$, the bigger c_j is, the more emergency the recipient is

w_{ij} : the matching level between donor i and recipient j

Where $w_{ij} \in [0,1]$, the bigger w_{ij} is, the more effective the match is

The objective function is:

$$\max \sum_{j=1}^B c_j \sum_{i=1}^A t_{ij} \cdot \frac{1}{d_{ij}} \cdot w_{ij}$$

Constraint is: $d_{ij} < d_{\max}$

Where $d_{\max}$ is the maximum distance between donor and recipient

Then the model is

$$\begin{aligned} \max \quad & \sum_{j=1}^B c_j \sum_{i=1}^A t_{ij} \cdot \frac{1}{d_{ij}} \cdot w_{ij} \\ \text{s.t.} \quad & \begin{cases} d_{ij} < d_{\max} & (i=1,2,\dots,A; j=1,2,\dots,B) \\ \sum_{j=1}^B t_{ij} \geq 1, & (i=1,2,\dots,A; j=1,2,\dots,B) \\ \sum_{i=1}^A t_{ij} \geq 1, & (i=1,2,\dots,A; j=1,2,\dots,B) \\ 0 \leq c_j \leq 1, & (j=1,2,\dots,B) \\ 0 \leq w_{ij} \leq 1 & (i=1,2,\dots,A; j=1,2,\dots,B) \\ t_{ij} = \begin{cases} 1, & x_i \text{ is matched with } y_j \\ 0, & \text{otherwise} \end{cases} \end{cases} \end{aligned}$$

Algorithm 1

The problem is a maximum bipartite matching problem, we solve it with maximum bipartite matching algorithm^[1].

A bipartite graph $G = (X \cup Y, E)$, where E is the edge set.

- Initialize M to any arbitrary (possibly empty) matching.
- While there exists a free x -vertex and a free y -vertex
- Let r be a free x -vertex. Using breadth-first search, grow an alternating path tree T rooted at r .
- If T is a Hungarian tree then let $G \leftarrow G - T$ {remove T }
- else find an augmenting path p in T and let $M = M \oplus p$
- end while

Model 2 Optimize allocation of resources model (OARM)

There is a distance between the donate center and the transplant place (generally, it is a hospital). Assume there are m donate centers and n transplant places in a range (nation or region), a_i is the location of donate center i , b_j is the location of organ transplant center j .

r_{ij} is the distance from a_i to b_j , s_{ij} is the supply from a_i to b_j ($i=1,2,\dots,m; j=1,2,\dots,n$), l_{ij} is the weight from a_i to b_j , including the distance between donate center to transplant center, the emergency of the patient in transplant center j . $l_{ij} \in [0,1]$, the bigger l_{ij} is, the higher priority it have.

$$\begin{aligned} \max \quad & \sum_{i=1}^m \sum_{j=1}^n s_{ij} \cdot l_{ij} \\ \text{s.t.} \quad & \begin{cases} r_{ij} < r_{\max} & (i=1,2,\dots,m; j=1,2,\dots,n) \\ \sum_{j=1}^n s_{ij} = u_i, & (i=1,2,\dots,m) \\ \sum_{i=1}^m s_{ij} = v_j, & (j=1,2,\dots,n) \\ 0 \leq l_{ij} \leq 1 \end{cases} \end{aligned}$$

Solution to the model

The objective of the model is a maximum objective. Firstly, we change the objective function into its equivalent objective function, and the model become

$$\begin{aligned} \min \quad & \sum_{i=1}^m \sum_{j=1}^n s_{ij} \cdot \frac{1}{l_{ij}} \\ \text{s.t.} \quad & \begin{cases} r_{ij} < r_{\max} & (i=1,2,\dots,m; j=1,2,\dots,n) \\ \sum_{j=1}^n s_{ij} = u_i, & (i=1,2,\dots,m) \\ \sum_{i=1}^m s_{ij} = v_j, & (j=1,2,\dots,n) \\ 0 \leq l_{ij} \leq 1 \end{cases} \end{aligned}$$

Then the problem is change into a problem similar to transportation problem.

Algorithm 2: table dispatching method (TDM)

Table 1

Sales area manufacturing location	B_1	B_2	$L \ L \ L$	B_n	output
A_1	S_{11} L_{11}	S_{12} L_{12}	$L \ L \ L$	S_{1n} L_{1n}	U_1
A_2	S_{21} L_{21}	S_{22} L_{22}	$L \ L \ L$	S_{2n} L_{2n}	U_2
M M M	M M M	M M M	$L \ L \ L$	M M M	M M M
A_m	S_{m1} L_{m1}	S_{m2} L_{m2}	$L \ L \ L$	S_{mn} L_{mn}	U_m
sale	V_1	V_2	$L \ L \ L$	V_n	

The solution to transportation problem accord to the theorem of simplex method and the feature of transportation problem, and design a method can compute on the table. The main steps of the algorithm are:

Step 1: get a basic feasible solution

Step 2: check whether the current basic feasible solution is the optimal solution or not.

If so, stop the iteration, or turn to the next step.

Step 3: develop the current basic feasible solution, and get a new basic feasible solution, turn to step 2.

Concrete steps of the algorithm

1. get the basic feasible solution with minimum element method

step 1: set $I = \{1, 2, \dots, m\}$, $J = \{1, 2, \dots, n\}$, $\Delta = \emptyset$, $x_{ij} = 0 (i = 1, 2, \dots, m; j = 1, 2, \dots, n)$;

step 2: decide values of p and q , $c_{pq} = \min\{c_{ij} \mid i \in I, j \in J\}$;

step 3: $\varepsilon = \min\{a_p, b_q\}$, set $x_{pq} = \varepsilon$ and fill into the cell at the crosspoint of A_p

and B_q and cycled. And $a_p = a_p - \varepsilon$, $b_q = b_q - \varepsilon$, $\Delta = \Delta \cup \{x_{pq}\}$, if $a_p = 0$ then

$I = I - \{p\}$, if $b_q = 0$ then $J = J - \{q\}$.

Step 4: Check whether $a_p + b_q = 0$ or not. If $a_p + b_q = 0$ then turn to step 5, or turn to step 2.

Step 5: check whether $I = \emptyset$, if $I = \emptyset$ then $x_{ij} (i=1,2,\dots,m; j=1,2,\dots,n)$ is the solution to the problem, Δ is the basic variable sets and stop the algorithm, or $c_{rs} = \min \{c_{ij} \mid i \in I, j \in J\}$, $\Delta = \Delta \cup \{x_{rs}\}$, set $p = r, q = s$ and turn to step 3.

The m constraints are $u_1, u_2, \dots, u_m$, the latter n constraints are $v_1, v_2, \dots, v_n$.

2. Through the formula group

$$\begin{cases} u_i + v_j = c_{ij} \\ u_i = 0 \end{cases} \quad x_{ij} \in \Delta$$

We can get the potential energy(PE) or dual solution u_i and v_j ($i=1,2,\dots,m; j=1,2,\dots,n$) or compute on the table directly.

3. Compute the check number

$$\sigma_{ij} = c_{ij} - (u_i + v_j) \quad (i=1,2,\dots,m; j=1,2,\dots,n);$$

And then $\sigma_{rs} = \min \{\sigma_{ij} \mid 1 \leq i \leq m; 1 \leq j \leq n\}$

4. check whether $\sigma_{rs} \geq 0$ or not. If so then $x_{ij} (i=1,2,\dots,m; j=1,2,\dots,n)$ is the optimal solution and stop the algorithm. Or set off from x_{rs} and find a cycle in the table, and the vertex in the cycle all are the vertex in Δ . Numbering the vertexs in E from x_{rs} . The vertexs with the odd number are in the vertex set E^+ , and with the even number are in the vertex set E^- (it is obvious that $x_{rs} \in E^+$)

5. $\theta = \min \{x_{ij} \mid x_{ij} \in E^-\} = x_{kt}$

$$6. \quad x_{ij} = \begin{cases} x_{ij} + \theta, & \text{when } x_{ij} \in E^+ \\ x_{ij} - \theta, & \text{when } x_{ij} \in E^- \\ x_{ij}, & \text{otherwise.} \end{cases}$$

Set $\Delta = (\Delta \cup \{x_{rs}\}) - \{x_{kt}\}$ and return to step 2.

According to the objectives and constraints in model 1 and model 2, we can see that the main factors attributing to the potential bottlenecks are the shortage of donors

(especially the shortage of deceased donors), transplant technology and difficult difficult matching.

If more resources were available for improving the efficiency of the donor-matching process, consider the three bottlenecks mentioned above, we should use the increased resources on the three bottlenecks. So the concret methods are:

- Because the shortage of donor is objective fact, it is not easy to solve the problem in a short period. A suggest is that the government can develop the policy of donation. For example, giving the donor and his/her relatives some right or priority in law or in medical to motivate the donation.
- Develop the technology to transplant some organ that is incompatible in current time, and then can enhance the efficiency of the transplant and increasing the success rate of the transplant.
- In the text, some algorithm is established to allocate the donor's organ reasonability and enhance the match efficiency.

If the network was divided into smaller networks (for instance at the state level), the distance between the donate center and transplant center is r_{ij} , and $r_{ij} \leq r_{\max}$ so the

model only need to consider the matching and organ optimize allocation of resources. It can be divided into two conditions:

- the organ resource in the state is abundant, and can satisfy the transplant needs of the state.
- The allocation among states are not fair and with bad harmony.

Conclusion

The smaller network can get improve the matching efficiency. For example, in a state with abundant organ, the time of the organ diliver from donate center to transplant center is much smaller than that in the nation. But it will cause some problem simultaneously. Patients in the state which lacks of organ resources is difficult to find a suitable organ to transplant and the life is threatened with death, while the organs in the state with abundant organs are leaved unused. It is a waste of the organ resources.

So the smaller network may have a good efficiency in small number of states, but to most of the states, it is with low efficiency.

4.2 State Laws^[4]

The widespread uniformity among key state laws is not a result of chance. In 1968, the National Conference of Commissioners on Uniform State Laws (NCCUSL) disseminated the Uniform Anatomical Gift Act. The 1968 Act offers a few main clarifications of how the donation process might occur. It declares who has the right to keep and to use a dead body. It proclaims that any eighteen year old with a sound mind may donate his or her body after death to be used for medical research or as a source of transplantable tissues and organs. The Act also defines the priority list of those who might donate the body if the person who died did not declare a preference for or against donation. By 1972, all fifty states and the District of Columbia had adopted the 1968 Act.

In 1987, the NCCUSL revisited the issue with the intention of increasing participation in organ donation. It revised the 1968 Act in a few main ways. First, the 1987 Act explicitly states that the declaration of a gift does not require any witnesses, whereas the 1968 Act required witnesses for the gift to be valid. Second, the document of a gift is sufficient for the removal of organs, which means the procurement team does not need the consent of the spouse or the family in addition to the document of gift. Third, the 1987 Act requires that hospital and medical personnel ask patients upon admittance if they would consider being organ donors. This practice is commonly called “routine inquiry.” Fourth, when a person who is dead did not officially declare an interest or a lack of interest in donating his or her organs upon death, hospital or medical personnel are required to discuss with that person’s relatives or surrogates whether or not they would be interested in donating that person’s organs. This practice is commonly referred to as “required request.” Fifth, the 1987 Act permits the removal of organs by medical personnel without explicit consent from a potential donor or from a relative of the donor, so long as the appropriate medical personnel or authorities have made a “reasonable effort” to discover any objection by the donor or the donor’s family. Finally, the 1987 Act forbids the buying and selling of organs, which reiterates the federal ban enacted in 1984. Since the promulgation of the 1987 Act, twenty-four states have adopted it.

In general, most states have similar laws regarding organ donation, as a few examples will demonstrate. Every state recognizes the driver’s license as a document of gift. Every state claims there are two loci of consent, the individual person and that person’s family. Every state declares that an eighteen year old can consent without needing parental permission. Every state has some form of an organ donation education program. Most states have a first person consent law, which means the document of gift is sufficient evidence for proceeding with the procurement of organs. Most states have donor registries. Either the Department of Motor Vehicles (DMV) provides the local OPO with donor information collected from driver’s licenses or the OPO collects the information itself. Finally, most states subscribe to the same ways of determining death. These states base their definitions on the NCCUSL’s Uniform Determination of Death Act.

That said, there are a few exceptions to the uniformity of state laws on organ donation. The location of the donor laws within the different state codes is one exception worth noting. Thirty-four states categorize their anatomical gift laws under a title that includes the word “health,” but the remainder of the states put their donor laws under a variety of titles. Louisiana uses the title “Education.” Florida uses the title “Civil Rights.” Colorado and Kentucky put their laws under a title that includes the word “professions.” New Hampshire, Washington, and Wisconsin have their laws under titles that include the phrase “human remains” or “dead bodies.” Alaska, Georgia, Illinois, Indiana, Maryland, Minnesota, Montana, Ohio, Oregon, and Pennsylvania encode their laws under a title that includes the words “estates,” “probate,” or “property.” It might be worth considering what each of these categories implies about decedents, their worth, and their relation to the living.

Another exception to the uniformity of state laws on organ donation is the choice of some states to have anatomical gift funds. Nineteen states currently have gift funds for increasing public awareness about the need for organ donation. Ohio, which has some of the most donor friendly laws of all the states, uses its fund not only to promote awareness of need but also to support local and statewide programs that publicly recognize donor families for the gift their deceased relative has given. In addition, Kentucky, Maine, and New York each dedicate a day or a week to publicly recognize organ donors for their sacrifice. These are the only states that have such programs, though.

There is a final major exception to the uniformity of state laws on organ donation. A few states have laws that aim to limit or remove the disincentives potentially faced by living organ donors. A disincentive here means any financial cost the donor might incur during the donation process. A few states choose to give state employees paid leave while they donate organs. The leave varies from two days in Colorado; to thirty days in Delaware, D.C., Maryland, Missouri, Utah, and Virginia; to six weeks in Ohio. While Arkansas does not grant state employees paid leave, it requires that private employers must grant unpaid leave of absence for organ donation, and that if the employer grants paid leave then the employer is entitled to a tax credit of 25% of the regular salary paid to the employee. Georgia law also grants a tax deduction, but directly to donors for un-reimbursed donation costs of up to \$10,000. Iowa uses 50% of its donor awareness trust fund to cover the costs of living donors, recipients, and transplant candidates.

As the previous account demonstrates, state laws primarily cover the process of donation, from the education of potential donors, to the consent requirements to be a donor, to donor registries, and finally to determining when a potential donor has died. While state laws focus mainly on the donation process, federal laws and regulations establish the boundaries of acceptable practice for the procurement, the allocation, and the transplant

4.3 Development of model

To prolong the life of the patients and saving more lives, the current transplantation policy need be improved.

The demand for organs for transplantation drastically exceed the number of donors currently, and there are many transplant candidates waiting for a organ transplantation in each state and this number is increased in a high speed, the waiting list will longer and longer in the future. In this condition, it is significant to prolong and save the life of people by improve the organ transplantation allocating policy. In the development of the model, the main consideration is how to use the organ reasonable when several candidates are waiting for the same organ and save the maximum life overall. The allocation problem of organ mainly include the following situations:

- The organ resources is limited, but sometimes the organ may be occupied for a long time by a recipient who didn't need the organ urgently.

- Attribute to some factors, some organs were not used in time.
- The influence to organ allocation from recipient's age and the contribution to the society in the future

In the following part, considering the situations mentioned above, a reasonable and efficient organ allocation algorithm will be established to improve the organ allocation policies.

Assume there are n candidates waiting for the same organ in waiting list, the problem is how to select a final recipient from the n candidates. Before solve the problem, it is needed to decide the rules to select the final recipient, the candidates should obey the rules to get the priority to receive the organ. These rules mainly include:

- Turning-round time is short. Generally, the turning-round time span is a important rule to evaluate the efficiency of organ transplant system and to select the allocate method to find the final recipient. The turning-round time span is the time span from the candidate register to require a organ to the end of successful transplant.

To each recipient, they all want to shortest their waiting time in waiting list. But as a supervisor of the organ transplant system, he want to shortest the average turning –round time, because it would not only save the time and cost on organ transplant of the system but also save more lives and make most of the candidate feel satisfaction. The average turning-round time is

$$T = \frac{1}{n} \left[\sum_{i=1}^n T_i \right]$$

- Consideration of deadline. The deadline is the latest time that the organ have to transplant to the recipient. If the system can't make sure of this, then the waiting time of the candidate exceed the time that she/he can wait, the candidate will die. This rule is to make the number of candidate who receive the organ transplantation before deadline as much as possible.
- Rule of priority. In the organ transplant system, all candidate should obey the rule of priority so as to make sure the most organ needed candidate get the chance to organ transplant. To consider to priority of candidate, the following factors need to be considered.
 - The candidate is in urgency need of organ transplant.
 - The people who will do more contribution to the society in the future will considered in higher priority.
 - Children are with higher priority to acquire organ transplant than old people.

Considered the three rules mentioned above and combine with reality, the allocation policy can divide into two modes

1. Non-preemptive Mode

This mode is the simplest mode in the allocation methods. In this mode, once a organ is decided to transplant to a recipient, then no matter how urgent the other candidate is and no matter when the decided recipient would conduct the organ transplant, the organ would keep for the decided recipient. Other recipient

can only wait for the next matching organ. This mode is mainly used in first come first served algorithm. The mode is easy to conduct, and it is good for the candidate who apply for the organ in the earliest time, but it is adverse to urgency candidates, and may cause death to many other candidate who can't acquire the organ transplant in time. In addition, the average turning-round time of the mode is very long, so the mode is not good.

2. Preemptive Mode

In this allocation mode, it is allowed to reallocate the organ to another candidate after the organ has already allocate to one candidate according to some rules.

The rules that needed to obey in reallocation are:

- Highest priority first(HPF)
- Highest response ratio first(HRRF)
- Time slice rule

Firstly, consider the first rule Highest priority first. According to the influence factors mentioned above, the priority can given to the candidates as follows:

- Urgency candidate first: The candidate that in urgent need of organ transplant will grant high priority. Then when the urgent candidate come to the system, as his/her priority is high than the others, the organs that have allocated to candidates but have not yet transplant to the candidate, then the organ that match the new coming patient would be reallocate to the new candidate in urgent need of organ transplant. So she/he would acquire the organ transplant in the shortest time.

This rule can not only shorten the average turning-round time of the system, but also save more lives. But this rule is adverse to the candidate whose disease is not so serious, she/he may in a long time of waiting, and it is unfair to him/she personally.

- The people with more contribution to the society in the future first. This is a rule under dispute because it is disobey the ethical rule of the organ transplant. But personally, the rule can maximum the value of the society.
- Younger people first: The people with small age was grant higher priority than the people with big age. For example, child have the highest priority to acquire the organ transplant.

3. Improve to the algortithm above: Highest response ratio first(HRRF)

In Urgency candidate first algorithm, it is performed well from the system. The main disadvantage of the algorithm is that the benefit of the not so serious diseased candidate is not well protected. If we import the dynamic priority to each candidate, and the priority of each candidate increase when the waiting time increase. Then after waiting for some time, the not so serious diseased candidate can get a high priority and acquire the chance to organ transplant. The dynamic change of the priority can be described as followed:

$$\text{Priority} = \frac{\text{waitting time} + \text{urgency of the disease}}{\text{urgency of the disease}}$$

It can be seen from the upper equation that:

- If the waiting time of two candidate are the same, then the more urgency the disease is, the higher priority she/he will have, so it is good for urgent diseased candidate.
- If the urgency of the two candidate's disease are the same, then the priority is decided by the waiting time. The longer the candidate wait, the higher priority he will granted.
- For the unserious diseased candidate, the priority can increase when the waiting time is increase. When she/he has wait for enough time, the priority will reach to a high point to let she/he get the chance of organ transplant.

In conclusion, the algorithm not only considered the benefit of urgent candidate, but also the benefit of not so urgent candidate for it considered the waiting time of the candidate. It is a good middle course. Of course, in this algorithm, before each allocation, it is needed to compute the response ratio. It will increase the complexity of the algorithm.

4. Algorithm based on Time slice rule

The algorithms mentioned above may be influenced by some subjective factors. For example, when decide the urgency of the disease, it is all attribute to the subjective decision. So we consider the following situation: A candidate has been granted a organ for transplant. But for some reason, she/he couldn't go to organ transplant on time, but the organ is still keep for she/him, while some other candidates are waiting for the same organ. So the current system is unreasonable in this case, and the candidates who not are allocated to a matching organ will unsatisfy with the allocate policy.

Based on the situation mentioned above, in the following part algorithm based on time slice rule is imported.

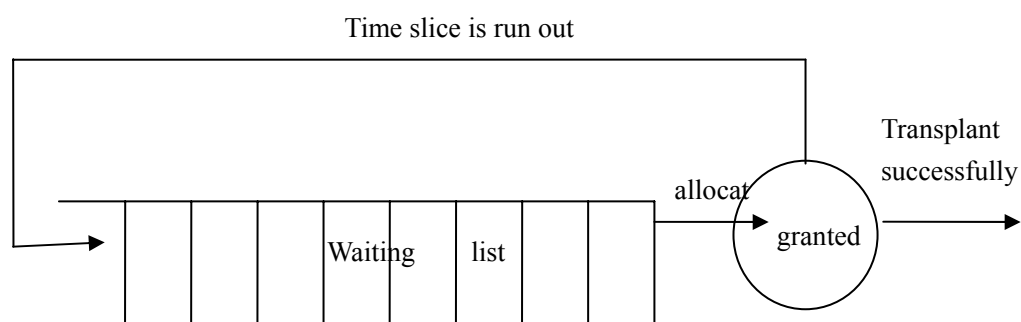


Fig.5

The main theory of the algorithm are: All candidates are granted a time slice of the same size. The time slice can be one week, one month, one year or more than one year (one time slice is smaller than the survival time of organ). The time slice granted to one candidate is the period that the organ can only transplant to this candidate. But if she/he didn't use the organ granted to she/him in the time slice, the organ will granted to another matching candidate for another time slice.

This algorithm can make sure that there are no organ was granted to one candidate for a long time that it can't used to others who need the same organ. It can improve the efficient of the transplant system. In addition, to improve the efficient of the system futhur, we can consider the time slice to be different among all candidates. The time slice is decide by some factors. It can make every candidate have the chance to get the matching organ in a efficient way.

5 Task 2

5.1 Analysis

To investigate the transplantation policies used in a each country, we first discuss the policies used in US, and then analyze the policies used in other countries such as Spain, Sweden, Iran and China.

1. The Current Policy^[4]

There are four main kinds of rules governing the organ transplantation process: state laws, federal laws, federal regulations, and United Network for Organ Sharing (UNOS) policies. The state laws primarily cover issues pertaining to the donation process, such as the criteria for declaring death, the consent requirements for being a donor, the scope of public education programs, and the composition of donor registries. With a few notable exceptions, these laws do not differ significantly from state to state. The federal laws primarily outline the processes of organ procurement, allocation, and transplantation by establishing the Organ Procurement and Transplantation Network (OPTN) and guidelines for organ procurement organizations (OPO). OPTN is the national organization that coordinates, implements, and monitors the organ transplantation system as a whole; OPOs are local or regional organizations responsible for retrieving organs and notifying potential recipients. The federal law also sets boundaries on how organs might be transferred from one person to another by prohibiting the buying and selling of organs. The federal regulations expand on the federal laws by describing how the OPTN, OPOs, and transplant hospitals should function. More specifically, they explain the internal structure of the organizations and the nature of their principal operations. UNOS rules further define the functions of the OPTN, and describe the policies OPOs and transplant hospitals must follow in order to be members of the OPTN. Collectively, these laws, regulations, and rules affect every group and every individual involved or potentially involved in the donation, procurement, allocation, and transplantation of organs. This part describes in more detail the most significant state laws, federal laws, federal regulations, and UNOS policies.

Alternative Policies in Other Countries

1. Presumed Consent in Spain

Spain is widely perceived in the transplant community as having one of the best organ donation programs in the world, and many analysts believe that its presumed consent law accounts for part of its success. As mentioned above, presumed consent in the organ donation context means that the government and procurement organizations may assume that citizens are willing to donate their organs at death if they do not state otherwise. There are two main kinds of presumed consent laws:

“Strong” presumed consent laws rely solely on the individual citizen to declare himself or herself a non-donor. “Weak” presumed consent laws allow the family members of a decedent to opt-out if they desire to do so. Spain has a weak presumed consent law, where hospital staff members always approach the surviving family members—not to request permission to procure organs, but to see if they would prefer not to allow procurement to proceed as it normally would.

2. Sweden’s Law of Mandatory Choice

In the spring of 1996 the Swedish health authority, Socialstyrelsen, distributed enough pamphlets about organ donation to reach Sweden’s entire population. The pamphlets specifically requested to know whether or not each individual would like to donate his or her organs at death. In the summer of that year Sweden began implementing a law that required each citizen to decide whether or not they wish to donate, and what was a request in the spring became a command in the summer. This statute is commonly referred to as Sweden’s “mandatory choice” law. We cannot say what the precise effect of this law has been, but we do know that Sweden ranks relatively low in donation rates compared to other Western countries.

3. Iran’s Organ Market

Iran’s transplant program is one of the most active in the Middle East. Unlike most other countries in the world, the vast majority of organs come from non-related living donors. And unlike in most other countries, many of these donors receive payment for their body parts—typically kidneys—which are legally bought and sold. In Iran, donors are actually vendors. The Charity Association for the Support of Kidney Patients and the Charity Foundation for Special Diseases are the two nongovernmental organizations that organize and monitor the organ trade. They match donors to recipients, and provide the donors with a specific sum of money for the donation. Without this source of kidneys from living donors, the organ supply would probably drastically diminish, in part because of significant disagreement among Islamic scholars over the guidelines for cadaveric donation.

4. Organ Conscription in China

In July 2006, China banned the buying and selling of human organs. Presumably this ban might affect the Chinese policy of routinely harvesting organs from prisoners, or at least those organs that the government is selling. The policy of organ conscription has its roots in the 1983 “Strike Hard” campaign, which announced that China would begin executing common criminals. With this new supply of decedents, China then created the “Rules Concerning the Utilization of Corpses or Organs from Corpses from Executed Criminals” law the following year. This law claims that the government can use a prisoner’s organs if that prisoner consents to donate. In November 2005, at a World Health Organization meeting, the vice minister of China’s Ministry of Health disclosed that 95% of organs transplanted in hospitals in China come from executed criminals, with the other 5% coming from living donors. The first percentage is shockingly high, and given China’s human rights record one might rightly doubt that most prisoners who surrender their organs freely consent to donation. This doubt explains why many organ transplantation scholars refer to the Chinese system as organ conscription.

It is showed that the transplantation policies in US is typical of the policies in most of the countries in the world. The policies of these countryies are similar to the policy in US. Of course, there are some policies different from the US. Policy in some countries such as the countries mentioned above. Take Iran as a example, the US policy is not applicable to Iran. So it is needed to modify the model built in task 1. The model of US transplantation policy is:

$$\begin{aligned} \max \quad & \sum_{j=1}^B c_j \sum_{i=1}^A t_{ij} \cdot \frac{1}{d_{ij}} \cdot w_{ij} \\ \text{s.t.} \quad & \begin{cases} d_{ij} < d_{\max} & (i=1,2,\dots,A; j=1,2,\dots,B) \\ \sum_{j=1}^B t_{ij} \geq 1, & (i=1,2,\dots,A; j=1,2,\dots,B) \\ \sum_{i=1}^A t_{ij} \geq 1, & (i=1,2,\dots,A; j=1,2,\dots,B) \\ 0 \leq c_j \leq 1, & (j=1,2,\dots,B) \\ 0 \leq w_{ij} \leq 1 & (i=1,2,\dots,A; j=1,2,\dots,B) \\ t_{ij} = \begin{cases} 1, & x_i \text{ is matched with } y_j \\ 0, & \text{otherwise} \end{cases} \end{cases} \end{aligned}$$

But in Iran, it is legal to trade organ in the market. So it is more complexity to solve the problem and we should consider more factors. Assume p_i is the lowest price of people i to his/her organ, and q_i is the highest price the recipient who need the organ can afford. Then the model turned to be model 3.

5.2 Model 3 Matching model based on price of organ

$$\begin{aligned} \max \quad & \sum_{j=1}^B c_j \sum_{i=1}^A t_{ij} \cdot \frac{1}{d_{ij}} \cdot w_{ij} \\ \text{s.t.} \quad & \begin{cases} d_{ij} < d_{\max} & (i=1,2,\dots,A; j=1,2,\dots,B) \\ \sum_{j=1}^B t_{ij} \geq 1, & (i=1,2,\dots,A; j=1,2,\dots,B) \\ \sum_{i=1}^A t_{ij} \geq 1, & (i=1,2,\dots,A; j=1,2,\dots,B) \\ 0 \leq c_j \leq 1, & (j=1,2,\dots,B) \\ 0 \leq w_{ij} \leq 1 & (i=1,2,\dots,A; j=1,2,\dots,B) \\ t_{ij} = \begin{cases} 1, & x_i \text{ is matched with } y_j \\ 0, & \text{otherwise} \end{cases} \\ p_i \leq q_i \end{cases} \end{aligned}$$

But in many situation, the price given by seller is higher than the price given by buyer: $p_i \geq q_i$. So individual requirements of both buyer and seller should be

considered. We import a concept named pricing tolerance: $\alpha_i = \frac{p_i - q_i}{p_i}$, $\alpha_{\max}$ is the

limit of the pricing tolerance. In reality, the buyer would lower the price as much as possible, but the seller have his/her limit of the lowest price. So set $\alpha_{\max}$ is the extreme case of the buyer and seller. Then the model can developed base on this theory.

5.3 Model 4: matching model based on pricing tolerance

$$\begin{aligned} \max \quad & \sum_{j=1}^B c_j \sum_{i=1}^A t_{ij} \cdot \frac{1}{d_{ij}} \cdot w_{ij} \\ \text{s.t.} \quad & \begin{cases} d_{ij} < d_{\max} & (i=1,2,\dots,A; j=1,2,\dots,B) \\ \sum_{j=1}^B t_{ij} \geq 1, & (i=1,2,\dots,A; j=1,2,\dots,B) \\ \sum_{i=1}^A t_{ij} \geq 1, & (i=1,2,\dots,A; j=1,2,\dots,B) \\ 0 \leq c_j \leq 1, & (j=1,2,\dots,B) \\ 0 \leq w_{ij} \leq 1 & (i=1,2,\dots,A; j=1,2,\dots,B) \\ t_{ij} = \begin{cases} 1, & x_i \text{ is matched with } y_j \\ 0, & \text{otherwise} \end{cases} \\ \alpha_i = \frac{p_i - q_i}{p_i} \leq \alpha_{\max}, & \text{where } \alpha_{\max} > 0 \end{cases} \end{aligned}$$

In conclusion, the model of the transplantation policy used in US is not suitable to the policy used in Iran, and need to develop the model based on the model to the policy in US. Similaly, the model need to modify while it is applied to the transplantation policy in Spain, Sweden and China. The public in Spain are of the high conscious to donate organ, and the Spanish government publicise the donation in a large scale. The Swedish government adopt the coercion policy in organ transplant. This policy is a way to increase the number of the organ. In China, it is banned to buy or sell organ. So it is necessary to build the suitable model to solve the problem in each country.

Table 2: The policies in the US and other countries

country	policy	features	Referable or not to the US	Method to reference
US	Optn net	Open, synthetical	/	
Spain	Presumed Consent	High propaganda power	Yes	Increase the propaganda power
Sweden	Mandatory Choice	Compulsoryiness	No	/
Iran	Organ Market	Allow to trade organ	No	/
China	Conscription	Organ is not allowed to trade, increase the donation conscious of people	Yes	Ban the organ trade, and increase the donation conscious of people

Accord to Table 2,the US can refer to the polices in Spain and China and increase the propaganda power. We import the propaganda power C , $C \in [1, \infty]$. The bigger the propaganda power is, the larger the number of the donation is. Then the improved model is:

$$\begin{aligned} \max \quad & \sum_{i=1}^m \sum_{j=1}^n s_{ij} \cdot l_{ij} \\ \text{s.t.} \quad & \begin{cases} r_{ij} < r_{\max} & (i=1,2,\dots,m; j=1,2,\dots,n) \\ \sum_{j=1}^n s_{ij} = u_i \cdot C, & (i=1,2,\dots,m) \\ \sum_{i=1}^m s_{ij} = v_j, & (j=1,2,\dots,n) \\ 0 \leq l_{ij} \leq 1 \end{cases} \end{aligned}$$

So by increasing the propaganda power, the number of the donors would be increased and then solve the problem of the bottlenecks for efficient organ matching.

5.4 Report to Congress

As the demand in organs for transplantation drastically exceed the number of donors currently, there are many candidate recipients waiting for a organ transplantation in each state and this number is increased at a high speed, the waiting list will become longer and longer in the future. So it is a big issue to solve the problems existed in organ transplant.

Through the anlysis of the given data, we can conclude that the main factor attributing to the increasing number of transplant candidates is the shortage of donors, and the potential bottlenecks for efficient organ matching are shortage of donors, organ transplant technology and difficulty in matching. So the key to solve organ transplant problem are to increase the resourse of the donations, to improve the transplant technology and find a suitable way of organ matching to enhance the matching probability.

To increase the resourse of the donations, it is important to enlarge the propaganda power to change the consciousness of people to the organ transplantation and to motivate people to donate organ. The bigger the propaganda power is, the larger the number of the donation is. It may also go well if some rights and priority given in law or in medical is granted to the donors or their relatives to motivate people donation.

Technology is another important factor that influence the organ transplant. It is needed to pay more attention to the research of the organ transplant to improve the technology of organ transplant.

To maximize the number of matching organ from several donors, the factors such as matching level, urgency of transplants of patients and the distance from donor to recipient should be considered. In addition, dividing the original network into smaller network can also improve the matching efficiency. For example, in a state with abundant organ, the time of the organ delivery from donate center to transplant center is much smaller than that in the nation. But it will cause some problem simultaneously. Patients in the state which is lack of organ resourses is difficult to find a suitable organ to transplant and the life is threatened with death, while the organs in the state with excessive organs are left unused. It is a waste of the organ resourses. So in our opinion, maybe the smaller network has a good efficiency in states of small number, but to most of the states, it is with low efficiency.

At the same time, it is significant to prolong and save the life of people by improving the organ transplantation allocation policy. By investigation in the transplanation policies used in a country other than the US, the allocation in different countries can not refer to the same allocation method because there are differences among their policies. The methods used to American policies should be modified when it is used in other country to adapt to the policies in the country.

6 Task 3

6.1 Background

Transplantation is the preferred treatment for the most serious forms of kidney disease. As of this writing there are over 68,000 patients on the waiting list for cadaver kidneys in the United States. By way of comparison, in 2006 there were over 16,000 transplants of cadaver kidneys performed in the United States.

Table 3: US Kidney Transplants. The data for years 1988-2006 are constructed from the annual report of UNOS-OPTN, Number of registrations may have multiple counts of patients since one patient may have registered in multiple centers for the waitlist.

years	1988	1989	1990	1991	1992	1993	1994	1995	1996
Cadaveric Donors	3874	3808	4304	4268	4276	4609	4794	5001	5036
Cadaveric Transplants	7061	6753	7322	7281	7204	7508	7639	7694	7730
Live Donors	1812	1903	2094	2396	2534	2851	3009	3392	3678
Live Donor Transplants	1812	1903	2094	2396	2534	2851	3007	3387	3668
Total Waitlist Patients	/	/	/	/	/	/	/	32392	33223
New Waitlist Registrations	/	/	/	/	/	/	/	17888	18306

1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
5083	5339	5386	5489	5528	5638	5753	6325	6700	6609
7774	8032	8043	8124	8230	8539	8667	9357	9913	9807
3933	4421	4724	5493	6038	6240	6473	6647	6570	5913
3927	4419	4717	5488	6035	6240	6470	6647	6568	5914
34678	36880	41209	43266	43798	46283	47705	52020	56410	57172
19048	20167	22784	23420	23434	24613	25584	28308	30335	31239

As these numbers make clear, there is a considerable shortage of cadaveric kidneys, compared to the demand. (We note in passing that the substantial consensus in the medical community remains firmly opposed to allowing organs—even cadaveric organs—to be bought and sold, and in most states this is a felony.) Congress views cadaveric kidneys offered for transplantation as a national resource, and the National Organ Transplant Act of 1984 established the Organ Procurement and Transplantation Network (OPTN). Run by the United Network for Organ Sharing (UNOS), it has developed a centralized priority mechanism for the allocation of cadaveric kidneys.

In addition to transplants of cadaver kidneys, in 2006 there were also somewhat over 6,600 transplants of kidneys from living donors, a number that has been increasing steadily from year to year. Transplants from live donors generally have a higher chance of success than those from cadavers. The way such transplants are typically arranged is that a patient identifies a healthy willing donor (a spouse, for example) and, if the transplant is feasible on medical grounds (i.e. appropriate blood types, and absence of positive crossmatch. antibodies), it is carried out. If the transplant from the willing donor is not feasible, the patient typically enters (or remains on) the queue for a cadaver kidney, while the donor returns home.

Constraint

1. Blood Type Compatibility

- There are four blood types: A, B, AB and O.
- In the absence of other complications:

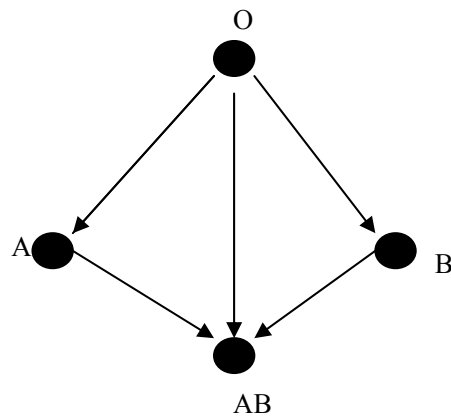


Fig.6

- Type O kidneys can be transplanted into any patient;
- Type A kidneys can be transplanted into type A or type AB patients;
- Type B kidneys can be transplanted into type B or type AB patients; and
- Type AB kidneys can only be transplanted into type AB patients.

2. Tissue Type Compatibility

- Tissue type or HLA type: Combination of six proteins.
- Prior to transplantation, the potential recipient is tested for the presence of preformed antibodies against donor HLA. If present (*positive crossmatch*), the transplantation cannot be carried out.

Paired Exchange and Indirect Exchange

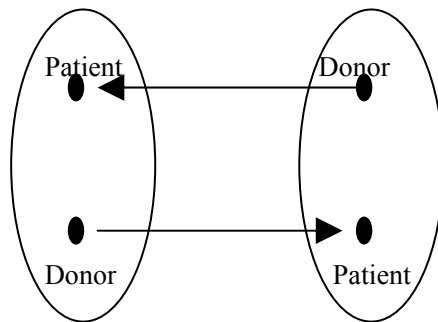


Fig.7 Paired exchange

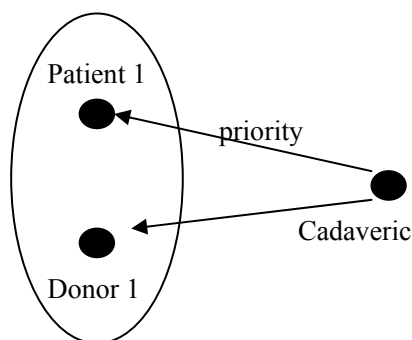


Fig.8 Indirect Exchange

Widespread concern in transplantation community: Indirect exchanges can harm type O patients with no living donors.

Nevertheless, many transplant centers have started pilot indirect exchange programs.

Building on the existing practices in kidney transplantation, we consider how an efficient and incentive-compatible system of exchanges might be organized, and what its welfare implications might be.

6.2 Model 5: Kidney Matching for Transplantation

Assumptions

- The graft survival rate increases as the tissue type mismatch decreases.
- There is no constraint on the number of transplants that can be simultaneously carried out.
- Indirect exchanges are feasible.

House Allocation^[5]

Shapley and Scarf modeled a housing market as consisting of n agents each of whom is endowed with an indivisible good, a house. Each agent has preferences over all the houses (including his own), and there is no money in the market, trade is feasible only in houses. They attribute to David Gale the top trading cycle algorithm that, for any preferences, produces an allocation of houses to agents in the core of the market. The algorithm can be described as follows: Each agent points to her most

preferred house (and each house points to its owner). Since the number of agents is finite and since each house has an owner, there is at least one cycle in the resulting directed graph. In each such cycle, the corresponding trades are carried out, i.e. each agent in the cycle receives the house she is pointing to, and these agents and houses are removed from the market. (A cycle may of course consist of an agent pointing to her own house.) The process continues (with each agent pointing to her most preferred house that remains on the market) until no agents and houses remain, and the final allocation is the one in which each agent receives the house with which she left the market.

Existing tenants are not only entitled to keep their current houses but also apply for other houses.

Algorithm 3: random serial-dictatorship with squatting rights (RSD-SR) mechanism^[6]

Step of algorithm RSD-SR:

- Step 1. Each existing tenant decides whether he will enter the housing lottery or keep his current house. Those who prefer keeping their houses are assigned their houses. All other houses become available for re-allocation together with the vacant houses.
- Step 2. An ordering of participating students (i.e. newcomers and existing tenants who decide to go ahead with the lottery) is randomly chosen.
- Step 3. Once the agents are ordered, available houses are allocated using the induced serial dictatorship: The first student is assigned his top choice, the next student is assigned his top choice among the remaining houses, and so on.

Analysis of algorithm 1

While this mechanism is common in applications, it suffers from a major deficiency. Since it does not guarantee each existing tenant a house that is as good as his own, some existing tenants may choose to keep their houses even though they wish to move, and this may result in loss of potentially large gains from trade.

Algorithm 4: top trading cycles (TTC) mechanism

The inefficiency of the RSD-SR is caused by the mechanism's failure to guarantee each existing tenant a house at least as good as the one he already holds. To achieve efficiency this deficiency needs to be addressed. So this algorithm is designed to solve the problem.

Step of algorithm top trading cycles (TTC)

- Step 1. Each student (existing tenant or newcomer) reports his strict preferences over all houses.
- Step 2. An ordering of agents is randomly chosen.
- Step 3. For any given preference list and ordering, the outcome is obtained using the following algorithm:
 - (a) Assign the first student his top choice, the second student his top choice among the remaining houses, and so on, until someone requests the house of an existing tenant.
 - (b) If at that point the existing tenant whose house is requested is already assigned another house, then do not disturb the procedure. Otherwise

modify the remainder of the ordering by moving the existing tenant to the beginning of the line and proceed with the procedure.

- (c) Similarly, move any existing tenant who has not already been assigned a house to the beginning of the line once his house is requested.
- (d) If at any point a cycle forms, it is formed by exclusively existing tenants and each of them requests the house of the tenant who is next in the cycle.

(A cycle is an ordered list of students $(i_1, i_2, \dots, i_k)$ where student i_1 requests the house of student i_2 , student i_2 requests the house of student i_3 , . . . ,

student i_k requests the house of student i_1 .) In such cases remove all students in the cycle by assigning them the houses they request and proceed with the procedure.

Algorithm 5: TTCC (top trading cycles and chains) Mechanism^[6]

Symbols

(k_i, t_i) : A donor – patient pair.

K_i : Living donor kidneys compatible with patient t_i .

ω : Priority in the waitlist in exchange for a live kidney.

P_i : Strict preferences over $K_i \cup \{k_i, \omega\}$.

Cycles and Chains

- A kidney exchange mechanism is a systematic procedure to select a matching for each kidney exchange problem.
- Top Trading Cycles and Chains mechanism relies on an algorithm consisting of several rounds. In each round:
 - each patient t_i points either towards a kidney in $K_i \cup \{k_i\}$ or towards ω
 - each kidney k_i points to its paired recipient t_i .

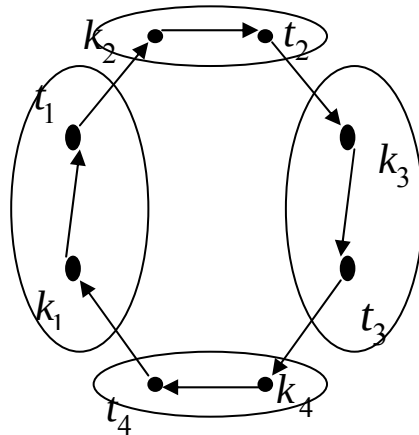


Fig.9 Cycles

Cycles are associated with direct exchanges. No two cycles can intersect.

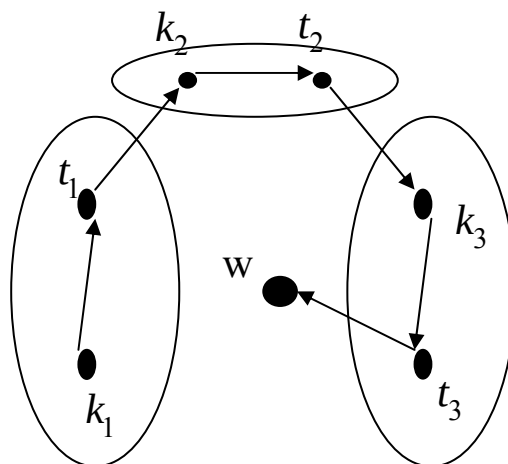


Fig.10 w-chains

w-chains are associated with indirect exchanges. w-chains can intersect!

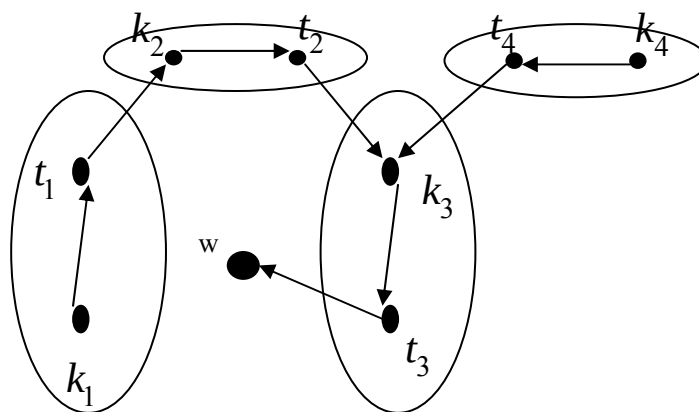


Fig.11 w-chains with intersect

- Unlike in a cycle, a kidney-patient pair can be part of several w-chains.
- Practical Possibility: Choosing among w-chains with a chain selection rule.
- Remark: Choice of the chain selection rule has efficiency and incentive-compatibility implications.

Lemma 1 . Consider a graph in which both the patient and the kidney of each pair are distinct nodes as is the waitlist option w . Suppose each patient points either towards a kidney or w , and each kidney points to its paired recipient. Then either there exists a cycle or each pair initiates a w-chain.

TTCC Mechanism algorithm

Fix a chain selection rule. The TTCC mechanism determines the exchanges as follows:

1. Initially all kidneys are available and all agents are active. At each stage
 - each remaining *active* patient t_i points to the best remaining unassigned kidney or to the waitlist option w , whichever is more preferred,
 - each remaining *passive* patient continues to point to his assignment, and

- each remaining kidney k_i points to its paired recipient t_i .
2. By Lemma 1, there is either a cycle, or a w-chain, or both.
 - Proceed to Step 3 if there are no cycles. Otherwise locate each cycle and carry out the corresponding exchange. Remove all patients in a cycle together with their assignments.
 - Each remaining patient points to its top choice among remaining choices and each kidney points to its paired recipient. Proceed to Step 3 if there are no cycles. Otherwise locate all cycles, carry out the corresponding exchanges, and remove them.

Repeat this step until no cycle exists.
 3. If there are no pairs left, then we are done. Otherwise by Lemma 1, each remaining pair initiates a w-chain.

Select only one of the chains with the chain selection rule. The assignment is final for the patients in the selected w-chain. In addition to selecting a w-chain, the chain selection rule also determines

 - whether the selected w-chain is removed, or
 - the selected w-chain remains in the procedure although each patient in it is passive henceforth.
 4. Each time a w-chain is selected, a new series of cycles may form. Repeat Steps 2 and 3 with the remaining active patients and unassigned kidneys until no patient is left.

Plausible Chain Selection Rules

Depending on policy priorities, one may consider adopting a number of alternative chain selection rules. We next give several examples:

- a. Choose minimal w-chains and remove them.

For this simplest chain selection rule the outcome is not affected whether all minimal w-chains are removed simultaneously or they are removed one at a time.
- b. Choose the longest w-chain and remove it. If the longest w-chain is not unique then use a tie-breaker to choose among them. One possibility is priority ordering patient-donor pairs based on factors such as waiting time and breaking the ties based on this priority order.
- c. Choose the longest w-chain and keep it. If the longest w-chain is not unique, choose one based on a tie-breaker.
- d. Prioritize patient-donor pairs in a single list. Choose the w-chain starting with the highest priority pair and remove it. This chain selection rule may be used to increase the inflow of O blood-type living kidneys to the cadaveric waiting list by giving higher priorities to pairs with O blood-type donors.
- e. Prioritize patient-donor pairs in a single list. Choose the w-chain starting with the highest priority pair and keep it.
- f. Prioritize the patient-donor pairs so that pairs with O blood-type donor have higher priorities than those who do not. Choose the w-chain starting with the highest priority pair; remove it in case the pair has an O blood-type donor but keep it otherwise.

Efficiency and Incentives

- Theorem 1 . Consider a chain selection rule where any w-chain selected at a non-terminal round remains in the procedure and thus the kidney at its tail remains available for the next round. The TTCC mechanism, implemented with any such chain selection rule, is efficient.
- Theorem 2 . Consider the priority based chain selection rules c and d. The TTCC mechanism, implemented with either of these chain selection rules is strategy-proof.
- Corollary. The TTCC mechanism, implemented chain selection rule d is efficient and strategy-proof.

6.3 Simulations

The theoretical treatment of the TTCC mechanism makes clear that larger exchanges may yield welfare gains, but it gives us no idea of their magnitude, i.e. whether they are worth pursuing. The following simulations are meant as a first step in that direction. We use data where it exists, e.g. on the likelihood of mismatches and positive crossmatches.

Where no data exists on the willingness of patients and donors to trade a live donation for priority on the cadaver queue we do robustness checks by simulating a wide range of preferences.

Table 4: ABO blood-type, gender, and age distribution for patients used in the simulations.

Patient Blood type	Frequency
O	45.6%
A	39.5%
B	11.1%
AB	3.8%

Patient Gender	Frequency
Female	40.9%
Male	59.1%

Patient Age	Frequency
<18	5.6%
18-34	13%
35-49	34.9%
50-64	38.9%
>64	7.6%

Table 5: Relation among Unrelated Donors, Age, and Positive Crossmatch Distributions for Living Donors.

Unrelated Living Donor	Frequency
Spouse	53.5%
Other	46.5%

Living Donor Age	Frequency
<18	5.6%
18-34	13%
35-49	34.9%
50-64	38.9%
>64	7.6%

Positive Crossmatch	Frequency
Female Patient-Husband	33.3%
Other	11.1%

Preference Construction

Using this analysis, and assuming that a patient's utility from a graft is a monotone decreasing function of its relative risk of failure determined by

- the age of the living-donor and
- the HLA mismatch number,

we derive the preferences of patients. we assume that the graft failure risk of a

patient from a live organ transplant is in functional form $a^x b^{\frac{y}{10}}$ where x is the number of HLA proteins in A, B and DR loci of the donor that do not exist in the recipient, also known as the HLA mismatch number (changes in the range $\{0, 1, 2, \dots, 6\}$), y is the age of the living-donor (changing in the open interval $(18, 80)$), a is the relative graft failure risk per HLA mismatch, and b is the relative graft failure risk per 10 years of increase in the donor age. Therefore, since a patient's utility over grafts is a monotone decreasing function of its relative risk of failure, the utility function

$$u(x, y) = -x \ln(a) - \frac{y}{10} \ln(b)$$

Result of the Simulations

In our simulations, we randomly simulate a sample of n donor-patient pairs using the population characteristics explained above. Then, we determine the preferences of patients over kidneys in the sample: For each patient t_i , we first check whether a donor k_j is ABO-compatible. If k_j is ABO-compatible, then we check whether there is a positive crossmatch between t_i and k_j . If they test negative for crossmatch, then k_j is in the compatible donor set K_i of patient t_i . After finding the set of compatible kidneys for each patient, we obtain a preference ordering on this set, using the utility functions described above. We construct 8 sets of preferences for each patient using the rational or cautious preference construction methods discussed

above and assuming that 0%, 10%, 20%, of patients with incompatible donors consider the waitlist option acceptable. We simulate each of mechanisms under these eight preference scenarios. We use a Monte-Carlo simulation size of 100 trials for three different population sizes of 30, 100, and 300 pairs.

Table 6: Number of Transplants and Quality of Match for n=30

	Mechanism	Total Trans.%	Waitlist Upgrade %	Rel.Risk of Failure	HLA
All	No Exchange	54.83	0	1	4.79
All	RSD-SR	68.5	0	1	4.78
Rational	TTC	82.47	0	0.95	4.16
Cautious	TTC	81.07	0	0.96	4.29
Waitlist 10%					
All	RSD-SR	68.5	2.8	1	4.78
Rational	TTCC	82.97	1.3	0.95	4.16
Cautious	TTCC	81.63	1.3	0.96	4.28
Waitlist 20%					
All	RSD-SR	68.5	6.17	1	4.78
Rational	TTCC	83.6	3.23	0.95	4.16
Cautious	TTCC	82.3	3.10	0.96	4.29

Table 7: Number of Transplants and Quality of Match for n=100

	Mechanism	Total Trans.%	Waitlist Upgrade %	Rel.Risk of Failure	HLA
All	No Exchange	54.79	0	1	4.83
All	RSD-SR	73.59	0	1	4.82
Rational	TTC	87.85	0	0.93	3.72
Cautious	TTC	87.23	0	0.93	3.86
Waitlist 10%					
All	RSD-SR	73.59	2.31	1	4.82
Rational	TTCC	88.47	1.05	0.93	3.72
Cautious	TTCC	87.86	1.22	0.94	3.87
Waitlist 20%					
All	RSD-SR	73.59	4.94	1	4.82
Rational	TTCC	88.81	2.05	0.93	3.72
Cautious	TTCC	88.37	2.57	0.93	3.87

Table 8: Number of Transplants and Quality of Match for n=300

	Mechanism	Total Trans.%	Waitlist Upgrade %	Rel.Risk of Failure	HLA
All	No Exchange	53.92	0	1	4.81
All	RSD-SR	75.03	0	1	4.81
Rational	TTC	91.05	0	0.91	3.29
Cautious	TTC	90.86	0	0.91	3.40
Waitlist 10%					
All	RSD-SR	75.03	2.49	1	4.81
Rational	TTCC	91.57	0.79	0.91	3.29
Cautious	TTCC	91.41	0.84	0.92	3.40
Waitlist 20%					
All	RSD-SR	75.03	4.96	1	4.81
Rational	TTCC	91.92	1.52	0.91	3.29
Cautious	TTCC	91.76	1.62	0.91	3.40

6.4 Conclusion

It is showed from Table 6- Table 8 that TTCC Mechanism can maximum the number of organ transplant considering the medical and psychological of the situation. The transplant ratio can reach about 90%. It is also can see the resulting influence to the waiting list from the table.

7 Task 4

7.1 Analysis

In this task, it is needed to give a strategy to patient to decide whether to take an offered cadaver kidney or to participate in a kidney exchange. Suppose a barely compatible-in terms of HLA mismatches-kidney become available from the cadaver queue, the problem is should the patients take it or wait for a better match from the cadaver or from an exchange. As the cadaver kidney has a short half-life than a live donor kidney, the patients need considered the blood and HLA match of the cadaver kidney, the condition of themselves and the risks to transplant of not to transplant. In addition, the probability to find a better match kidney is less and less as time goes on.. So the patients should consider all of these factors synthetically and make a decision. The problem is a dynamic programming problem. The patients want to maximize the benefit of themselves through a suitable decision.

Model design

T is the time interval from the time that a patient was able to get a kidney but she/he decide to wait for a better match kidney to the time that the patient died because of no better matching kidney or recovered because of a better matching kidney is transplant to she/him in the future. Divide T into n equivalent time intervals. t_i

is time interval i . Besides, assume λ_i is the pain in time interval i , ξ_i is the condition of patients in time interval i . As time goes on, the pain is increasing and the condition of the patient will more and more severe. Assume that the pain in time

interval i is related to the pain in the former time interval $i-1$, and the ratio between them is

$$\alpha_i = \frac{\lambda_i}{\lambda_{i-1}}$$

β_i is the risk of patient after transplantation, γ_i is the number of available kidney to the patient in time interval i , p_i is the probability to find a better match kidney in the future, m_i is the matching degree.

7.2 Model 6: Dynamic optimal matching decision model

$$\begin{aligned} & \max_{i=1, \dots, n} m_i \\ & \text{s.t.} \begin{cases} \sum_{i=1}^n t_i = T, i=1, 2, \dots, n \\ \alpha_i = \frac{\lambda_i}{\lambda_{i-1}}, i=1, 2, \dots, n \\ \lambda_{\min} \leq \lambda_i \leq \lambda_{\max}, i=1, 2, \dots, n \\ \beta_{\min} \leq \beta_i \leq \beta_{\max}, i=1, 2, \dots, n \\ \xi_{\min} \leq \xi_i \leq \xi_{\max}, i=1, 2, \dots, n \\ \gamma_{\min} \leq \gamma_i \leq \gamma_{\max}, i=1, 2, \dots, n \end{cases} \end{aligned}$$

To describe the matching effect m_i , m_i is expressed by λ_i , ξ_i , β_i and γ_i . As these four factors have different unit, so the weight vector $w=[w_1, w_2, w_3, w_4]$ is imported to standardize the four indicators, then the model can be changed into a multi objective synthetical planning model.

7.3 Model 7: multi objective synthetical planning model

$$\begin{aligned} & \max_{i=1, \dots, n} (w_1 \lambda_i + w_2 \xi_i + w_3 \beta_i + w_4 \gamma_i) \\ & \text{s.t.} \begin{cases} \sum_{i=1}^n t_i = T, i=1, 2, \dots, n \\ \alpha_i = \frac{\lambda_i}{\lambda_{i-1}}, i=1, 2, \dots, n \\ \lambda_{\min} \leq \lambda_i \leq \lambda_{\max}, i=1, 2, \dots, n \\ \beta_{\min} \leq \beta_i \leq \beta_{\max}, i=1, 2, \dots, n \\ \xi_{\min} \leq \xi_i \leq \xi_{\max}, i=1, 2, \dots, n \\ \gamma_{\min} \leq \gamma_i \leq \gamma_{\max}, i=1, 2, \dots, n \end{cases} \end{aligned}$$

Solution to the model

- Change the problem into multi step decision process.
- Select the suitable status variable s_k , which not only agree with the evolutive feature of the process, but also satisfy the Markovian feature.
- Decide the decision variable x_k and the decision set $D_k(s_k)$ in each time interval.
- Get the right state transition equation $s_{k+1} = T_k(s_k, x_k)$.
- Get the right target function V_{kn}

$$\text{Where } V_{kn} = \sum_{i=k}^n v_i(s_i, x_i) \quad (k=1, 2, \dots, n)$$

$v_i(s_i, x_i)$ is the target function in time interval i . The target function can also be described as

$$V_{kn} = v_k(s_k, x_k) + V_{k+1,n}(s_{k+1}, x_{k+1}, \dots, s_n, x_n).$$

Since the target function is the function of initial state and strategy, so the equation above can be written as

$$V_{kn}(s_k, p_{kn}(s_k)) = v_k(s_k, x_k) + V_{k+1,n}(s_{k+1}, p_{k+1,n}(s_{k+1})).$$

According to the equation above, the optimal function is

$$f_k(s_k) = \underset{p_{kn} \in D_{kn}(s_n)}{\text{opt}} V_{kn}(s_k, p_{kn}(s_k))$$

And

$$\begin{aligned} \underset{p_{kn} \in D_{kn}(s_k)}{\text{opt}} V_{kn}(s_k, p_{kn}(s_k)) &= \underset{\{x_k, p_{k+1,n}\}}{\text{opt}} \{v_k(s_k, x_k) + V_{k+1,n}(s_{k+1}, p_{k+1,n}(s_{k+1}))\} \\ &= \underset{x_k \in D_k(s_k)}{\text{opt}} \{v_k(s_k, x_k) + \underset{p_{k+1,n} \in D_{k+1,n}(s_{k+1})}{\text{opt}} \{s_{k+1}, p_{k+1,n}(s_{k+1})\}\}, \end{aligned}$$

while

$$f_{k+1}(s_{k+1}) = \underset{p_{k+1,n} \in D_{k+1,n}(s_{k+1})}{\text{opt}} V_{k+1,n}(s_{k+1}, p_{k+1,n}(s_{k+1})),$$

so

$$f_k(s_k) = \underset{x_k \in D_k(s_k)}{\text{opt}} \{v_k(s_k, x_k) + f_{k+1}(s_{k+1})\} \quad (k = n, n-1, \dots, 1).$$

To make the recursion process successfully, a bound constraint should be added to the model. It is

$$f_{n+1}(s_{n+1}) = 0$$

A group of basic equation in the following can be get from the recursion equation above:

$$\begin{cases} f_k(s_k) = \underset{x_k \in D_k(s_k)}{\text{opt}} \{v_k(s_k, x_k) + f_{k+1}(s_{k+1})\} & (k = n, n-1, \dots, 1) \\ f_{n+1}(s_{n+1}) = 0; \\ s_{k+1} = T_k(s_k, x_k) \end{cases}$$

The solution process is: Get the optimal decision and optimal value in each time interval by backward computation from $k = n$. At last, the algorithm will get the optimal value $f_1(s_1)$ of the whole procession. Compute $f_1(s_1)$ by input the value of s_1 in it. Then through the state transition equation, s_2 can be computed by s_1 and x_1^* . Then repeat the process and get x_n^* at last and then get the optimal decision

$$p_{1n}^* = \{x_1^*, x_2^*, \dots, x_n^*\}.$$

The latter part of the process above is called back substitution.

7.4 Model 8: Optimal decision model based on entropy and grey synthetical evaluation

Assume that the probability p_i of the occurrence of a better match kidney are the same in every time interval, $p_i = p$. As the number of considered factors is great, we just select some of the indicator as followed: pain degree λ_i , the condition ξ_i of the patient, risk β_i after transplant and the number γ_i of available organ.

Entropy technology

The entropy technology is a multi indicator decision method based on the conception of entropy in physics and chemistry. Its main theory is to depict the quality of a proposal by entropy which is used to describe the chaos, and change the multi indicator decision problem into single indicator decision problem efficiently.

Considered the decision matrix $D = \{x_{ij}\}$ of the multi indicator decision problem with m proposals (m time intervals) and n indicators.

Compute the characteristic weight of indicator j in proposal i , it is :

$$p_{ij} = \frac{x_{ij}}{\sum_{i=1}^m x_{ij}}$$

Then the output entropy value E_j of indicator j is:

$$E_j = -K \sum_{i=1}^m p_{ij} \ln p_{ij} \quad (j=1,2,\dots,n)$$

Where $K = 1/\ln(m)$

As $0 \leq p_{ij} \leq 1$, we can get $0 \leq -\sum_{i=1}^m p_{ij} \ln p_{ij} \leq \ln(m)$, then $0 \leq E_j \leq 1$ ($1 \leq j \leq n$)

Then the deviation is: $d_j = 1 - E_j$

If the decision make have no preference among the n indicators, then the n indicators are of the same preference. In this case, the weight w_j of indicator j is defined as:

$$w_j = \frac{d_j}{\sum_{j=1}^n d_j}$$

If the decision make have preference λ_j on indicator j , then λ_j can be improve by the weight w_j gained above and get a relatively accuracy estimate of the weight of each indicator:

$$\hat{\lambda}_j = \frac{\lambda_j w_j}{\sum_{j=1}^n \lambda_j w_j}$$

The step to solve the problem are as follows

Step 1. Decide the optimal indicator set (F^*)

$F^* = [j_1^*, j_2^*, \dots, j_n^*]$, where $j_k^* (k=1,2,\dots,n)$ is the optimal value of indicator k .

This optimal value can be the real optimal value in all proposals (If the indicator of bigger value is better, then it is the maximum value of the indicator among all proposals; if the indicator of smaller value is better, then it is the minimum value of the indicator among all proposals.), or it can be the accepted optimal value in public. When make decision of optimal value, it not only need to consider the advantage, but also should consider the feasibility. If the optimal value is too big to reach, then the result will be not accurate.

After decide the optimal indicator sets, matrix D can be constructed:

$$D = \begin{pmatrix} j_1^* & j_2^* \dots & j_n^* \\ j_1^1 & j_2^1 \dots & j_n^1 \\ M & M & M \\ j_1^m & j_2^m \dots & j_n^m \end{pmatrix}$$

Where j_k^i is the original value of indicator k in proposal i .

Step 2. Normative approach for the value of indicator

As the values of different kind of indicators have the different unit and order of magnitude, they can't be compared directly. To make sure the reliability of the result, it is needed to make normative approach to the original data.

Assume the range of the indicator k is $[j_{k1}, j_{k2}]$, where j_{k1} is the minimum value of indicator k in m proposals while j_{k2} is the maximum value. Then the original value can be changed into no unit value $C_k^i \in (0,1)$ with the formula following

$$C_k^i = \frac{j_k^i - j_{k1}}{j_{k2} - j_{k1}}, i = 1, 2, \dots, m; k = 1, 2, \dots, n$$

Then the matrix D is changed into matrix C :

$$C = \begin{pmatrix} C_1^* & C_2^* \dots & C_n^* \\ C_1^1 & C_2^1 \dots & C_n^1 \\ M & M & M \\ C_1^m & C_2^m \dots & C_n^m \end{pmatrix}$$

Step 3. Compute the result of overall evaluation

According to the theory of grey system, take $\{C^*\} = [C_1^*, C_2^*, \dots, C_n^*]$ as referenced series and $\{C\} = [C_1^i, C_2^i, \dots, C_n^i]$ as the compared series, then get the incidence coefficient $\xi_i(k)$ of indicator k to optimal indicator k in proposal i . It is:

$$\xi_i(k) = \frac{\min_i \min_k |C_k^* - C_k^i| + \rho \max_i \max_k |C_k^* - C_k^i|}{|C_k^* - C_k^i| + \rho \max_i \max_k |C_k^* - C_k^i|}$$

Where $\rho \in [0,1]$ and generally $\rho = 0.5$

Through $\xi_i(k)$, it is easy to get the evaluation matrix E :

$$E = \begin{pmatrix} \xi_1(1) & \xi_1(2) \dots & \xi_1(n) \\ \xi_2(1) & \xi_2(2) \dots & \xi_2(n) \\ M & M & M \\ \xi_m(1) & \xi_m(2) \dots & \xi_m(n) \end{pmatrix}$$

Where $\xi_i(k)$ is the incidence coefficient of indicator k to optimal indicator k

in proposal i.

Then the overall evaluate result is

$$R = E * W,$$

$$\text{Or } r_i = \sum_{k=1}^n W(k) * \xi_i(k)$$

Where W is the weight get by entropy technology.

If the correlative degree r_i is the maximum among all correlative degree, then n

$\{C^i\}$ is the closest indicator to the optimal indicator $\{C^*\}$. That is to say, proposal i is better than others. According to the value of correlative degree, it is easy to get the quality of each proposal.

Solution to the model

Firstly, analyze the original data and divide T into five time intervals.

Considered the fact that pain would increase along with as time goes on, so we assume that the value of λ_i indicate the pain of patient, and the pain increased more and more quickly as time goes on. and at last reach the limit of the pain. Then the value of λ_i can be

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \\ \lambda_5 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.3 \\ 0.5 \\ 0.8 \\ 0.9 \end{bmatrix}$$

By the same way, considering the condition of patients, the risk after transplant and the number of available kidney, we can get:

$$\begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \\ \xi_4 \\ \xi_5 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.3 \\ 0.5 \\ 0.8 \\ 0.9 \end{bmatrix} \quad \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.6 \\ 0.5 \\ 0.4 \\ 0.3 \end{bmatrix} \quad \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \\ \gamma_4 \\ \gamma_5 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

In which λ_i , ξ_i and β_i are minimizing indicators, while γ_i is maximizing indicator.

By the four vectors gained above, construct matrix A.

$$A = \begin{bmatrix} 0.2 & 0.2 & 0.8 & 5 \\ 0.3 & 0.3 & 0.6 & 4 \\ 0.5 & 0.5 & 0.5 & 3 \\ 0.8 & 0.8 & 0.4 & 2 \\ 0.9 & 0.9 & 0.3 & 1 \end{bmatrix}$$

According to the computing step of entropy technology, the non-preference weight can be gained by computing with MATLAB.

$$W_1 = [0.3029, 0.3029, 0.1229, 0.2714]$$

If the risk after transplant is the most important factors, then set the preference vector is

$$\tau = [0.2, 0.2, 0.4, 0.2]$$

Then the weight is change into:

$$W_2 = [0.2697, 0.2697, 0.2189, 0.2417]$$

As λ_i , ξ_i and β_i are minimizing indicators, γ_i is maximizing indicator, then the optimal indicator set is:

$$F^* = [j_1^*, j_2^*, \dots, j_n^*] = [0.2, 0.2, 0.3, 5]$$

Construct the matrix as followed:

$$D = \begin{bmatrix} 0.2 & 0.2 & 0.3 & 5 \\ 0.2 & 0.2 & 0.8 & 5 \\ 0.3 & 0.3 & 0.6 & 4 \\ 0.5 & 0.5 & 0.5 & 3 \\ 0.8 & 0.8 & 0.4 & 2 \\ 0.9 & 0.9 & 0.3 & 1 \end{bmatrix}$$

The non-preference result gained by grey synthetical evaluation technology is:

$$R_1 = E * W_1 = \begin{bmatrix} 0.9253 \\ 0.7237 \\ 0.5469 \\ 0.4344 \\ 0.4236 \end{bmatrix}$$

Then modify the result with preference, we get:

$$R_2 = E * W_2 = \begin{bmatrix} 0.8670 \\ 0.7012 \\ 0.5546 \\ 0.4704 \\ 0.4867 \end{bmatrix}$$

Table 9

	R_1	R_2
Time interval 1	0.9253	0.8670
Time interval 2	0.7237	0.7012
Time interval 3	0.5469	0.5546
Time interval 4	0.4344	0.4704
Time interval 5	0.4236	0.4867

7.5 Conclusion:

According to the table above, if patients have no preference on indicators, they would select to transplant right now. When patients have some preference on indicators, for example have preference on the risk after transplant, they should also select to transplant right now, but wait for a short time is also allowed because the weight in time interval 2 is close to the weight in time interval 1.

The accuracy of the result: According to the policies in most of the countries, the kidney should transplant within 1 day, especially the cadaver kidney, because it has a shorter half-life than a live donor kidney. Considered in medical perspective, the constraints of kidney transplant is not so strict. If there are more than three HLA match, the kidney is available to the patients, and it is allowed to transplant for more than one times. What's more, the kidney demands are much more than the supply. So it is better to transplant as soon as possible. It agree with the reality well.

8 Task 5

8.1 Some advice to the criteria and policies^[4]

A. The Expansion of Education Programs

These efforts could be significantly expanded. Numerous education programs are already in place, and while surely useful and valuable, it seems unlikely to many observers that more education alone will drastically change donation behavior.

B. Remove Disincentives for Living Donation

Disincentives are the financial costs a potential organ donor might incur during the process of living donation. Some of the most significant disincentives are: lost pay and forgone advancement due to missed work; the costs of transportation, from the initial visit to determine eligibility for donation to the final trip home after

transplantation and recovery; lodging and perhaps food during these trips; and any other financial costs directly related to the donation process. It aims to use public funds and public law to help altruistic individuals help others in need, yet it does so without undermining the system and spirit of organ “gifting.” It recognizes that giving an organ involves a grave personal sacrifice, and that we should not make donation to a loved one impossible for those who could not bear the burden of lost pay or the extra out-of-pocket expenses.

C. The Use of Expanded Criteria Donors

One way to expand the potential pool of organ donors is to loosen the health criteria for being a donor at death. There are two main ways of expanding this definition. First, the age limit might be increased to 75 or 80. Second, some of the health requirements imposed on potential donors might be reduced. Serious diseases such as AIDS would still disqualify the use of organs, but less severe conditions might not.

Expanding the criteria would require specific changes in the rules governing organ transplantation. There might, however, need to be some updates of the informed consent policies, letting recipients of older, less healthy organs know of the increased risks of receiving such organs. In addition, the rules of allocation might need to be altered to accommodate this new tier of organs.

D. The Promotion of Organ Exchanges

One possible way to expand the donor pool is to promote and facilitate organ exchanges. In some cases, an individual wishes to act as a living donor for a loved one, but cannot do so because he or she is not biologically compatible with the intended recipient. An organ exchange would permit the living donor to give his or her organ to someone else in exchange for the intended recipient receiving an organ from another source. There are two main forms of such exchanges: list donations and paired exchanges.

- List Donation

A list donation occurs when Donor A desires to give a kidney to Recipient A, but the two individuals are not biologically compatible. Instead, Donor A gives a kidney to someone on the general kidney list, and in exchange Recipient A receives a compatible kidney from the list. Depending on how such a policy is enacted, Recipient A could receive a certain number points to improve his or her list ranking, meaning he or she would get a kidney earlier but not immediately. Or the intended recipient could be moved up to the top of the list, meaning that he or she would get the next publicly available kidney.

- Paired Exchanges

A paired exchange is similar to list donation, but it involves two living donors instead of one living donor and a kidney from the list. In other words: Donor A gives to Intended Recipient B, and Donor B gives to Intended Recipient A. Paired exchanges can also occur among multiple pairs of donors and recipients. For example: the donor from pair A has a kidney that matches the recipient of pair B. The donor of pair B has a kidney that matches the recipient of pair C. And the donor of pair C has a kidney that matches the recipient of pair A. Each donor can then offer a kidney to a compatible recipient, allowing three donations to occur instead of (in some cases) none at all.

E. The Alteration of Consent Laws

This category includes a range of policy options that would modify or fundamentally alter the ways of consenting to organ donation. While any typology is an oversimplification, there seem to be five main kinds of consent laws: (1) first person consent, which is when one becomes an organ donor only by explicit personal consent; (2) family consent, which is when the surviving family members give explicit permission for a deceased loved one to be an organ donor; (3) mandatory choice, which is when the government requires every adult citizen to use his or her right to consent by declaring himself or herself either a donor or a non-donor; (4) presumed consent, which is when the government assumes that organs can be procured from the deceased unless the individual or the family expresses the explicit desire to “opt out”; and (5) organ conscription, which is when the government procures organs, from the dead or even the living, without the individual having any real choice to opt out.

F. The Legal and Clinical Redefinition of Death

At present, nearly every state sanctions two standards for declaring someone dead: a neurological standard (death of the whole brain, including the whole brain stem) and a cardiopulmonary standard (permanent cessation of heart and lung functions). Critics of current practice come from many different directions, but there seem to be four basic alternatives that one might consider as policy options:

- **Cardiopulmonary Definition:** Some argue that even those who meet the standard for whole brain death are not dead, or they argue that the condition of brain-dead individuals is too ambiguous to make a confident declaration of death. They propose, therefore, that all declaration of death be made according to cardiopulmonary criteria alone.
- **Higher Brain Definition:** Some argue that those who have lost the higher brain functions, even if they retain some autonomic brain function, are already dead, because they have lost all those qualities that are necessary for a distinctly human life. They propose, therefore, permitting the declaration of death by a more lenient higher-brain standard, meaning that some individuals

now considered alive would be considered dead and thus potentially deceased donors.

- The Self-Definition of Death: Some argue that because defining death is such a biologically and morally ambiguous matter, individuals should be permitted to declare, by means of a living will, which reasonable definition of death should apply to them—ranging from the strictest cardiopulmonary standard to the whole brain-death standard to the higher brain-death standard.
- Criteria for Dying Donors: Some argue that we should disentangle the question of defining death from the question of who is an eligible donor. This would mean developing clinical and legal criteria for permitting organ removal from the severely debilitated and/or the nearly dead—that is, that class of patients whose medical condition is defined as futile and/or for whom death is imminent. Rather than debating interminably the best definition of death, they argue that we should establish protocols for using dying donors.

G. New Rewards or Incentives for Organ Donation

One range of policy options involves creating new incentives or offering new rewards for being an organ donor. While disincentives are the costs that originate with the act of donation, incentives are the rewards offered to those who donate. These potential incentives vary significantly in type, and include: (1) public honors or medals for organ donors and/or their families; (2) gifts to charity in the name of the donor; (3) preferred status as a potential donor recipient; (4) health insurance coverage for living donors or insurance premium reductions for those who agree to donate their organs after death; (5) life and disability insurance for living donors; (6) payment of a deceased donor's funeral expenses; (7) cash payment for organs, with either fixed prices set and paid by the government or a true organ market with prices set by the laws of supply and demand. We will consider each of these options briefly in turn.

H. Promote Medical Alternatives to Transplantation

Most of the policy reforms considered above are animated by the desire to increase the organ supply and thus come to the aid of those now waiting for transplantable organs. Other policy proposals involve ensuring that donors, both living and deceased, are treated with dignity and never exposed to undue risk or insufficient respect. Another approach—not incompatible with current policy or with the reforms considered above—is to look for ways to decrease the *need* for organs by preventing organ decline or repairing organ failure without requiring human organs. All we can do here is highlight three notable strategies, each of them worthy of much greater consideration and analysis: (1) preventative medicine; (2) xenotransplantation; and (3) regenerative medicine.

Based on the recommended exchange procedure and patient strategy, some ethical issues are discussed in the following section. Then main ethical issues discussed include the selection of donor, the selection of the recipient and organ transplantation.

8.2 Ethical Issues

The ethical issues in a paired-kidney-exchange program include the general issues that pertain to any program involving living donors, as well as issues specific to a paired exchange of kidneys.

A. Benefits and Risks for Donors

The primary benefit for living donors is psychological. Even if the transplantations fail, donors know that they did everything possible to help their loved ones. Since the donation is indirect in a paired exchange, the psychological benefit may be more diffuse than that of a direct donation.

The main medical risk is that entailed by the operative procedure, and this risk is not affected by whether the donor is related or unrelated to the recipient. Recent surveys show a perioperative mortality rate of 3 deaths per 10,000 donors. Other major complications (such as pulmonary embolus) occur in less than 2 percent of cases.

Several studies have examined long-term morbidity after kidney donation; one found an increased risk of hypertension, and another did not. No risk of progressive renal failure has been reported, although some donors have nonprogressive mild proteinuria. Some donors experience depression or conflict with family members. In general, these problems are unrelated to the success of the transplantation, and virtually all donors state that if asked again, they would make the same decision. Nevertheless, donors may feel angry or guilty, particularly if there is an adverse outcome, and such feelings may be exacerbated by the fact that they do not know the results of their own donations.

B. Benefits and Risks for Recipients

The primary benefit for the recipient is the timely receipt of a healthy kidney, which may reduce or obviate the need for dialysis and may minimize suffering before transplantation.

The main risks for the recipient are those inherent in kidney transplantation. Although long-term survival is improved in most groups of patients with end-stage renal disease who receive transplants, as compared with those treated with dialysis, the mortality rate associated with transplantation may be higher in the short term, particularly among blacks. There are also the psychological costs of asking a close friend or relative to donate a kidney. With a paired kidney exchange, the imposition on

the donor may be perceived as greater, because the donor gives a kidney to a stranger rather than directly to a loved one. The recipient may feel burdened by a debt that he or she can never repay.

C. Coercion and Informed Consent

The decision to donate a kidney is encouraged because a loved one is seriously ill and needs a healthy kidney, although this need is tempered by the availability of dialysis, which makes it possible for patients with end-stage renal disease to live for many years without a transplant. In a study of the informed-consent process in living-donor organ programs, Fellner and Marshall found that most donors made immediate decisions to donate their kidneys, and additional information did not change their decisions. Others, however, have found that family members can refuse. Research has also shown that although some families pressure members to donate an organ, others oppose a family member's willingness to donate.

Concern about coercion has led a few organ centers to reject all living donors. A potential donor may feel a strong obligation to donate a kidney because of guilt, love, duty, or loyalty. The views of other family members may intensify these feelings, but familial influence is not necessarily tantamount to coercion. The need to balance selfishness and altruism is a universal feature of family relationships — indeed, of all human interactions — and is surely not unique to organ transplantation; conflicts between selfishness and altruism thus do not invalidate voluntary consent.

The hesitant donor is usually given many opportunities to withdraw consent. Concern about coercion may be heightened with paired kidney exchanges, because a reluctant donor cannot invoke ABO incompatibility as the reason for not proceeding with the donation. The psychiatric evaluation in the exchange protocol should help ensure that coercion is minimized and that the donor's decision is made voluntarily. Although we acknowledge the complex dynamics of donation, we hope that repeated interactions with the transplantation team will allow a reluctant donor the opportunity to decline.

Some recipients may decide that they cannot ask a potential donor to donate a kidney to a stranger, even though the risks of donation are unchanged and the donation permits the kidney exchange to occur. For this reason, the potential recipient needs to give informed consent before the potential donor undergoes screening for the exchange protocol.

D. The Right to Withdraw Consent

Obviously, donors must have the opportunity to change their minds. The only way to ensure that both recipients in a paired exchange receive their grafts (i.e., that neither donor withdraws from the exchange agreement) is to perform the two transplantations simultaneously. This practice eliminates the possibility that Donor A gives Recipient

B a kidney, only to have Donor B then decide not to give a kidney to Recipient A. Rapaport envisioned a coordinated program in which the participating hospitals remove the organs simultaneously and exchange them by special courier. Our protocol involves only one hospital, which may simplify the timing of the procedure and communication between the surgical teams.

E. Privacy and Confidentiality

Strict privacy and confidentiality should be maintained for each donor–recipient pair. Although the recipients may want to express their gratitude to the donors, there is the possibility of anger or frustration if one recipient (or donor) does not fare as well as the other recipient (or donor). Breaches of confidentiality may have unanticipated repercussions for all parties involved in the exchange, even those with good medical results.

If four operations are performed at the same hospital at the same time, it may be difficult to maintain privacy and confidentiality. Nevertheless, we will attempt to do so. We intend to use different operating suites for the two pairs of donors and recipients and to house them in different units of the hospital. Many health care workers will be involved in the care of the two donor–recipient pairs, and everyone involved will be instructed about the importance of protecting patient confidentiality.

F. Commercialization and Exploitation

Most countries have signed a World Health Organization statement condemning the practice of buying and selling organs. In the United States, only altruistic donations are permitted; federal law prohibits the commercial exchange of organs. In developing this proposal, we considered whether an exchange of organs between two donor–recipient pairs might be regarded as a "transfer [of a] human organ for valuable consideration," which is prohibited by U.S. law. In our view, this would be a misinterpretation of the intent of the law, which was to prevent the exploitation of living persons who might be willing to sell their body parts for profit. The law was not designed to proscribe altruistic donations of organs by family members or close friends. The exchange program enlists altruistic donors who, because of medical criteria (ABO incompatibility), cannot donate organs directly to the intended recipients but are willing to make donations that will benefit their loved ones indirectly.

G. Public Acceptance

A major concern with all novel transplantation proposals is the reaction of the public; much has been written about how different methods to increase the availability of organs (such as using cadavers without heartbeats or infants with anencephaly) would increase public distrust of transplantation in general. This concern is expressed about transplants from living as well as cadaveric donors, even though several studies

have found that public attitudes toward altruistic living donations are quite positive and that physicians are more distrustful of the use of living unrelated donors than the public at large. A survey has shown that many transplantation centers that are willing to accept organs from unrelated donors rarely suggest this option to patients.

Considered the ethical issues in the criteria and policies, rank order the criteria and then discuss the reasonability of them. We consider the problem from two aspects.

Firstly, the medical principles are considered. The criteria is judged by the medical staff. Before transplanation, it is needed to evaluate the situation of the patients systhetically, and consider the probable effect of the transplant. The medical priciple is the basic and fair principle in all principles.

Then it comes to the ethical principles. For example, the principle that the people with bigger contribution to the society should have more rights for return; the principle that accord to the future value of the patients; the principle of family role (the important of the patients in a family).

The third principle is the the personal and social dealing ability. The personal dealing ability is the ability of the patient to coordinate with the treatment. The social dealing ability is the living condition of the patient related to transplanation such as habitat, economic condition, the importance of the patient to the other people and so on.

There are also principles about way of act and emotional preference. From the utilitarian, the selection of patients should mostly considered the development of technology and probability of successful transplant. But from the humanitarian perspective, it is need to select the recipient by medical principle for the non medical principle is not reasonable.

So the order of the principle to select the recipient and to allocate the organ are as followed:

1. Optimal principle: When the patients' relative donate for the patient, the patient should be considered first to receive the organ. The current situation of the relative's donation is showed in the following graph:

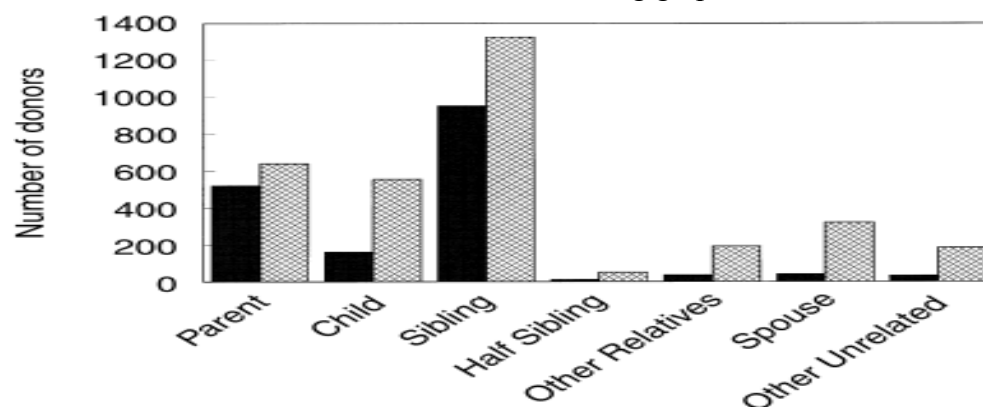


Fig.12 Relationship of living donors to recipients in 1988 (■) versus 2006(▨) showing the increase in distant relatives and unrelated donors.

2. Age principle: Considered the remaining life time and the quality of the life of the patients. Children under 18 years of age receive priority in waiting list. The current situation of the patient in each age interval is showed in the following table.

Table 10: The number of patient in each age interval(2006-10-27)

Age	The number of patients
< 18	748
18 to 34	8033
35 to 49	20553
50 to 64	28530
> 65	10628

3. Consider the contribution that the patient have do to the society.
4. Consider the future value of the patient.

8.3 Commercial renal transplantation

In some country, it is allowed to trade organs. However, in our opinion, it is harmful to trade organ in the long run though it can increase the number of the organ to a certain extent. The reasons that we object the organ trade are as followed:

1. The donor's choice is not voluntary because he is compelled by circumstances of poverty to donate a kidney
2. Paid donors usually are poor and uneducated, so making them understand the risks is all but impossible
3. Commercial donation will result in the rich having access to organs for transplantation while the poor do not.
4. Donors will be exploited by unscrupulous middlemen and surgeons. The medical care of both donor and recipient will suffer
5. The poor don't know how to handle the money that comes to them and it will make no permanent difference in their poverty
6. During its entire history, transplantation has relied on the altruism of donors and their families. Commercial donations would change the fundamental character of organ donation and likely would lead to the disappearance of altruistic donors. If any transplants are paid for, all will have to be.
7. The initial enthusiastic support of organ transplantation has been replaced by suspicion. Although no evidence has proved the charges, there are widely accepted urban myths regarding transplantation. These include the stories of people, particularly South American children, being kidnapped and killed for their organs,

and of people being drugged and kidnapped only to awaken in an alley with a flank incision and no kidney on that side.

8.4 A report to the director of the US Health Resources and Services administration

We think that there are some disadvantages in the policies and criteria in the US. Here are some suggestions for reference:

1. Remove the financial costs of the donors during the process of living donation. Some of the most significant financial costs are: lost pay and foregone opportunity of promotion due to missed work; the total cost of the transplantation used in transportation such as the cost of the initial visit to determine eligibility for donation, the cost of the final trip return back to home after the recovery of transplantation and so on. If the costs are not exist, much more people would likely to donation.
2. Promote the Organ Exchanges such as list donation and paired exchange for they have a good organ matching efficiency.
3. Give more rewards or incentives to organ donors. The incentives can include public honors or medals for organ donors or for their families; health insurance coverage for living donors or insurance premium reductions for those who agree to donate their organs after death and so on.
4. The legal and clinical redefinition of death: there are two standards for declaration of death. One is the neurological standard (death of the whole brain, including the whole brain stem) and the other is cardiopulmonary standard (permanent cessation of heart and lung functions). The neurological standard has more advantage to the transplantation, but it is objected by someone with ethical consideration. So a law with the redefinition of death is needed to solve the current state.

Considered the ethical problem in organ selection and transplantation, it include the following aspect of ethical problem:

1. Benefits and risks for donors .
2. Benefits and risks for recipients .
3. Forced and informed consent .
4. The right to withdraw consent .
5. Privacy and confidentiality .
6. Commercialization and exploitation .
7. Public acceptance .

The principle to select the recipient and to allocate the organ are as followed:

- Optimal principle: When the patient's relative donate for the patient, the patient should be considered first to receive the organ.
- Age principle: Considered the remaining life time and the quality of the life of the patients. Children under 18 years of age receive priority in the waiting list.
- Consider the contributions that the patient have done to the society.
- Consider the future value of the patient.

As to organ trade, in our opinion, it is harmful to sell organ in the long run though it can increase the number of the organ to a certain extent. The reasons why we object the organ trade are as follows: Firstly, maybe the donor's choice is not voluntary because he is compelled by circumstances of poverty to donate a kidney. Generally, the donors selling the organ are poor and uneducated, thus it is difficult to make them understand the risks of donation. Sometimes, the donors may be exploited by unscrupulous middlemen and surgeons, then the medical care of both donors and recipients will suffer. What is worse, the organ trade will cause that the poor can hardly get a available organ to transplant while the rich can acquire a matching organ easily.

9 Task 6

9.1 Section 1: The decision model of potential donor

Organ donate is a practical question influenced several factors. The donor faced some risk in the process of donation. The risks in volunteering involve assessing the probability of success for the recipient, the probability of survival of the donor, the probability of future health problems for the donor, the probability of future health risks (such as failure of the one remaining kidney), and the postoperative pain and recovery. These risks and some other factors such as the age of the donor, the economical situation of the donor's family will influence the decision of the donor directly. All these factors influenced the donor's decision directly.

In the following part, we discussed the effect of every factors separately.

Table 11. The factors affect the decision of the donor

Type of the factors	factors	Meaning of the factors
Independent variable	X_1	Probability of success for the recipient
	X_2	Probability of survival of the donor
	X_3	The probability of future health problem for the donor
	X_4	The probability of future health risks
	X_5	The post-operative pain and recovery
	X_6	Other factors (age, economic condition and so on)
Induced variable	P	Decision of the donor (whether to donate or not)

1. The probability of success for the recipient

It mainly refer to the probability that the recipient can transplant organ successfully. It affect the donor's decision directly.

Whether the organ transplant is successful or not is due to the external environment and inner factors of the recipient. For example, if the location of the recipient is far from the city, and of bad medical condition, the organ transplant will less possible to succeed due to the technology constraint. Another situation is that the recipient is in bad health when she/he was transplanted the organ. In this situation, the probability of success for the recipient is much less than she/he is in health condition.

If the probability of success for the recipient is too small, the potential donor would tend not to donate the organ because the risk is too high to donate the organ to the recipient in this situation. On the contrary, the donor would tend to donate her/his organ to the recipient because the organ would probable to save a life.

Based on investigation and material collected from the internet, we divide the

range of probability of success for recipient into seven levels, and a value is given to each level. In the first level, the probability is smaller than 50% and the value of the level is -3; in level 2, the probability is between 50% and 65% and the value of the level is -2; in level 3, the probability is between 65% and 75% and the value of the level is -1; in level 4, the probability is between 75% and 85% and the value of the level is 0; in level 5, the probability is between 85% and 90% and the value of the level is 1; in level 6, the probability is between 90% and 95% and the value of the level is 2; in level 7, the probability is bigger than 95% and the value of the level is 3. It can be expressed as the following table.

Table 12. The division to the probability of success for the recipient

the probability interval of the level	value
$\leq 50\%$	-3
50% – 65%	-2
65% – 75%	-1
75% – 85%	0
85% – 90%	1
90% – 95%	2
$\geq 95\%$	3

2. The probability of survival of the donor

It is a vital term to the donor and have significant influence to the donor's decision, because it is related to the problem whether the donor would die or not in the donation.

Consider the external environment and inner factors of the donors (such as age, healthy and so on). For example, if the transplant technology in the donor's location is in bad condition, the probability of survival of the donor would smaller than that donate in a place with advance technology. The health of the donor should also be considered, if the donor is weak, then he would survive in small probability. When the probability of survival is small enough, the donor would tend not to donate the organ.

Similar with the analysis of probability of success for the recipient, divide the range of probability of survival into 5 levels as showed in the following table, and a value is given to each level.

Table 13. The division to the probability of survival of the donor

Survival probability interval of the level	value
$\leq 60\%$	-2
60% – 75%	-1
75% – 85%	0
85% – 95%	1
$\geq 95\%$	2

3. The probability of future health problems for the donor

The health of the potential donor will be affected by the donation in the future. If it is adverse to the health after donate the organ, the potential donor would be anxious to donate the organ. If the donation has no bad effect to the donor, then the potential donor would likely to donate.

Based on investigation and material collected from the internet ,we divide the range of probability of future health problems for the donor into 6 levels as showed in the following table , and a value is given to each level.

Table 14. The division to the probability of future health problems for the donor

Future health probability interval of the level	value
$\leq 50\%$	-2
50% – 60%	-1
60% – 70%	0
70% – 80%	1
80% – 90%	2
$\geq 90\%$	3

4. The probability of future health risks (such as failure of the one remaining kidney)

After the donation, the donor may face some health risks. For example, the remaining kidney may failure as it burdened so much for a long time. So the potential donor must consider this kind of risk before donation, and it have some effect to the decision of donor. If the potential donor think this risk will be very high, then the

donor would tend to not donate the organ. The one don't take much attention to this kind of risk would more likely to donate the organ.

Based on invnvestigation and material collected from the internet, we divide the range of probability of probability of future health risks into 5 levels as showed in the following table, and a value is given to each level.

Table 15. The division to the probability of future health risks (such as failure of the one remaining kidney)

Future health risks probability interval of the level	value
$\leq 60\%$	-2
60% – 70%	-1
70% – 80%	0
80% – 90%	1
$\geq 90\%$	2

5. The postoperative pain and recovery

The postoperative pain and recovery are related to the quality of life in the future. So it is also one of the important factors the potential donor considered before decision. According to the data and investigation to the postoperative pain and recovery of donors who had already donate their organ, we divide the situation of postoperative pain and recovery into four situations. They are bad,common,good, perfect. Then they are valued by number -1, 0, 1, 2. That is to say, -1 means bad situation, 0 means common situation, 1 means good situation and 2 means perfect situation. It is showed in the following table, and a value is given to each level.

Table 16. The division to the postoperative pain and recovery

Situation of postoperative pain and recovery	value
bad	-1
common	0
good	1
perfect	2

6. Other factors

In addition to the risk factors mentioned above, the main other factor that affect the decision of the donor are age and the income. Generally, the conscious of donation become mature gradually along with the increase of age, and the probability of donation is increased. The income is also related to the donor's decision.

At the same time, the income is related to the age, and agree with the lifecycle theory.

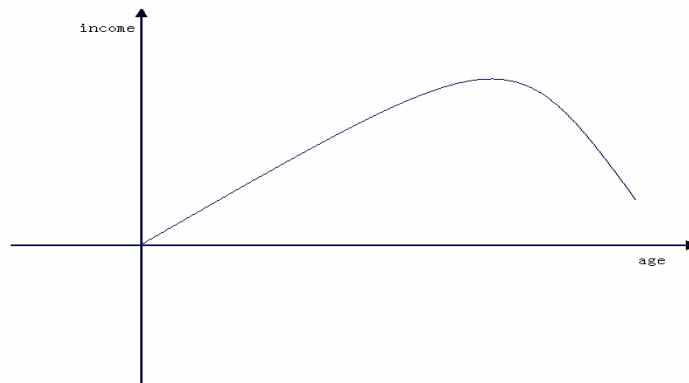


Fig.13

The income in one's youth and senectitude is much less than that in middle age. According to the relation between income and age, the two terms can be combined to one term by get the formula of the two terms. The new term can named to be weighted age.

Based on invnvestigation and material collected from the internet, the weighted age can be divided into 5 levels as showed in the following table, and a value is given to each level.

Table 17 division of weighted age

Weighted age interval in a level	level
≤ 30	-2
30 – 40	-1
40 – 50	0
50 – 60	1
≥ 60	2

Model 9: Logistic-regression model (LRM)

According to the influence from each factors and the value given to each level of each factors, the decision model of potential donors is established in the following part.

So the donor can decide whether to donate or not according to the value of the

factors. To model the donation decision, we use the logistic-regression^[11] model established by Pindyck R S and Rubinfeld D L, and then get different regression equations to the problem. The donor have two choice: donate her/his organ or not, it is the induced variable of the problem. The choice can be represent by 0 and 1 (0 means donate the organ and 1 otherwise).

The general form of logistic-regression model is

$$Y_i = \alpha - \beta X_i - \varepsilon_i$$

Where X_i is the value of the factors influence the decision of donor. Y_i represent the choice of donor:

$$Y_i = \begin{cases} 1 \\ 0 \end{cases}$$

Y_i satisfy the independent distribution and its value is always 0. The expected value of Y_i is

$$E(Y_i) = \alpha - \beta X_i$$

Since the value of Y_i is 0 or 1, so we can set the probability of the situation $Y_i=1$ is p_i , and the probability of $Y_i=0$ is $1 - p_i$, then the expected value is

$$\begin{aligned} E(Y_i) &= 1 \times p_i + 0 \times (1 - p_i) \\ &= p_i \end{aligned}$$

Considered the practical problem, the selected accuracy model of the problem is

$$P = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + L \quad L \quad + b_n X_n - \varepsilon$$

Where the value of P can only be 1 or 0.

$$P = \begin{cases} 1 & \text{the donor decide to donate the organ} \\ 0 & \text{the donor decide not to donate the organ} \end{cases}$$

X_i is the value of influence factors to donation, $i = 1, 2, 3, 4, L \quad L \quad, n$. n is the number of factors.

In general multiple regression, if P is the induced variable, then the equation is

$$P' = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + L \quad L \quad + b_n X_n$$

When compute P' by this equation, there sometimes exist the unreasonable situation such as $P' > 1$ or $P' < 0$, so the equation should change into the form likes

$$P' = \log_e \frac{P''}{1 - P''}$$

Then the Logistic regression equation is

$$P'' = \frac{e^{b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + L L + b_n X_n}}{1 + e^{b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + L L + b_n X_n}}$$

Through statistical analysis of the data with the help of SPSS and Logistic tools, a accuracy decision model can be established. Conduct model chi square test, block variable chi square test and error check to the model by Logistic regression to make sure the effectivity of the model.

In the study of the problem, we set six independent variable. They are the probability of success for the recipient, the probability of survival of the donor, the probability of future health problems for the donor, the probability of future health risks (such as failure of the one remaining kidney), and the postoperative pain and recovery. So $n = 6$. Then the donor decision model is

$$P = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + b_4 X_4 + b_5 X_5 + b_6 X_6 - \varepsilon$$

And

$$P'' = \frac{e^{b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + b_4 X_4 + b_5 X_5 + b_6 X_6}}{1 + e^{b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + b_4 X_4 + b_5 X_5 + b_6 X_6}}$$

Solve the model using two methods with the help of software SPSS.

1. All independent variables are forced to import to the regression equation. (METHOD = ENTER).
2. Conduct likelihood ratio test to the supposed parameters, and then decide the independent variables gradually. (/METHOD = FSTEP(COND)).

According to the material collected from the internet and some of the related data, through statistical analysis, the regression coefficient of the model are

$$\begin{aligned} b_0 &= 0.02 & b_1 &= 0.091 & b_2 &= 0.482 & b_3 &= 0.112 & b_4 &= 0.092 \\ b_5 &= 0.12 & b_6 &= 0.063 \end{aligned}$$

So the decision model of potential donor is

$$P' = 0.02 + 0.091X_1 + 0.482X_2 + 0.112X_3 + 0.092X_4 + 0.12X_5 + 0.063X_6$$

And

$$P'' = \frac{e^{0.02 + 0.091X_1 + 0.482X_2 + 0.112X_3 + 0.092X_4 + 0.12X_5 + 0.063X_6}}{1 + e^{0.02 + 0.091X_1 + 0.482X_2 + 0.112X_3 + 0.092X_4 + 0.12X_5 + 0.063X_6}}$$

According to the result of the significance test, all factors have remarkable influence to the decision of donors. Seen from the coefficients $b_0, b_1, \dots, b_6$

$$b_2 > b_3 > b_5 > b_4 > b_1 > b_6 > b_0$$

b_2 is the biggest one. So the probability of survival of the donor is the most important factor among all factors, and the probability of future health is the second most important factor. This conclusion agrees well with the reality. So increasing the probability of survival of the donor is the most important task to increase the number of donors, and the probability of future health is the next factors to be considered.

According to the decision model mentioned above, the donors can first determine in which level they are for each factor by considering their own situation. Secondly, get the value of each factor based on the level of each factor, and then compute the value of P . According to the value of P , the donor can make her/his decision.

Of course, the coefficient $b_0, b_1, \dots, b_6$ in the model are more or less influenced by the subjective opinion of the factors. For example, if one donor pays much attention to the post-operative of pain and recovery factor, then b_5 should be bigger than we get in the model, and the donor would go to the place with good reflection of this factor to donate the organ.

9.2 Section 2

In this section, the influence of perceived risks in transplant and personal issues to the decision of donor is discussed. As the influence factors are too much and some of them don't influence the decision of donor so much, we disregard some unimportant factors by using analytic hierarchy process (AHP) technology and just keep some of the important factors. Then the related department can accord to these factors to make policy so as to motivate the donation. In the following part, the analytic hierarchy process (AHP) technology is used to decide which factor should be paid attention to.

1. Construct the analytic hierarchy structure

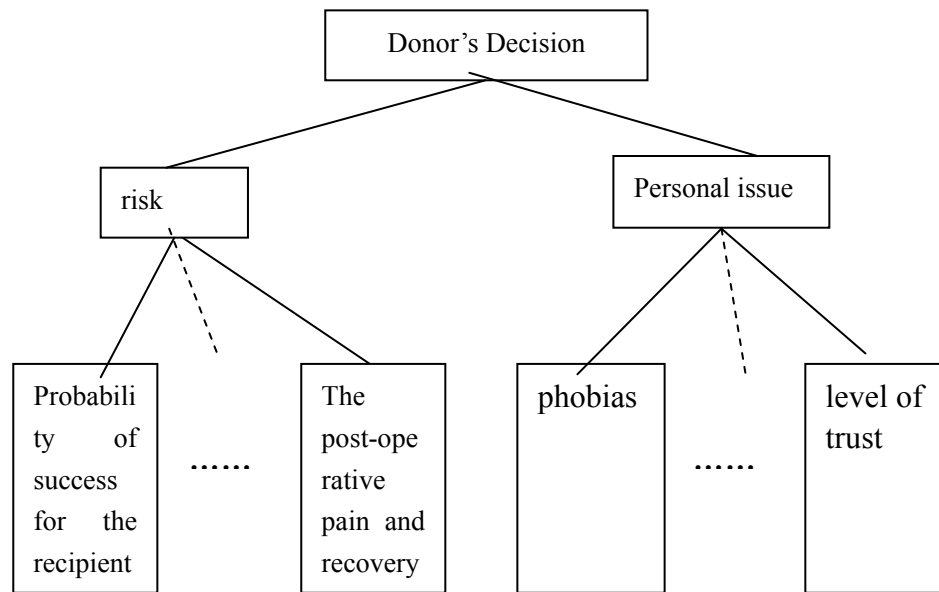


Fig.14

The top layer is the objective layer, the second layer is the two factors that affect the decision of donors, and the third layer is the concret factors of layer 2, in which risk factor include the probability of success for the recipient, the probability of survival of the donor, the probability of future health problems for the donor, the probability of future health risks (such as failure of the one remaining kidney), and the postoperative pain and recovery. The personal issue include phobias, irrational fears, misinformation, previous experiences with surgery, level of altruism, and level of trust.

According to the conclusion in the former part, the weight of the risk factors are b_1, b_2, b_3, b_4, b_5 . l_1, l_2, l_3, l_4, l_5 are the weight of personal issues that affect the decision of donors, and they can be calculated by the same method as computing the weight of risk factors. w_1, w_2 are the weight of the factors in layer two to the top layer, they are the proportion of the weight of risk and personal issue.

2. Construct the paired comparison matrix

Conduct pairwise comparison to the factors in each layer to construct the paired comparison matrix:

$$C = (C_{ij})_{n \times n}$$

Where C_{ij} is the ratio of the importance of factor i to the importance of factor j to the object.

Generally, C_{ij} have the following features:

$$(1) C_{ij} > 0$$

$$(2) C_{ij} = \frac{1}{C_{ji}} \quad \text{when } i \neq j$$

$$(3) C_{ii} = 1 \quad \text{when } i = 1, 2, 3, \dots, n$$

Matrix C is named positive anticlassic matrix. To the positive anticlassic matrix C , if $C_{ij} \cdot C_{jk} = C_{ik}$ to any i, j, k , then the matrix is called parity matrix.

Theorem of parity matrix:

- The rank of matrix C is 1, the single nonzero characteristic root of matrix C is n .
- Any column of C is the characteristic vector to characteristic root n .

If the paired comparison matrix is a parity matrix, the influence weight vector of the factors $C_1, C_2, C_3, \dots, C_n$ to the upper layer can be stated by the normalized characteristic vector related to the characteristic root. The vector is named weight vector.

If the paired comparison matrix is not a parity matrix, but the inconsistency is in the permissible range (that is to say the matrix can get through the consistency check), then the normalized characteristic vector of the maximum characteristic root λ can regard as the weight matrix.

$$C \cdot W = \lambda \cdot W$$

The concrete steps are as followed:

(1) Construct the paired comparison matrix of the third layer to the second layer

Based on the weight b_1, b_2, b_3, b_4, b_5 and l_1, l_2, l_3, l_4, l_5 , we can get the

following two matrixs C_1, C_2 :

$$C_1 = \begin{bmatrix} b_{11} & . & . & . & b_{1,5} \\ . & . & . & . & . \\ . & . & . & . & . \\ . & . & . & . & . \\ b_{5,1} & . & . & . & b_{5,5} \end{bmatrix} \quad C_2 = \begin{bmatrix} l_{11} & . & . & . & l_{1,5} \\ . & . & . & . & . \\ . & . & . & . & . \\ . & . & . & . & . \\ l_{5,1} & . & . & . & l_{5,5} \end{bmatrix}$$

Where b_{ij} and l_{ij} is the ratio of the two weight of factor i and j .

$$b_{ij} = \frac{b_i}{b_j} \quad i, j = 1, 2, 3, 4, 5 \quad l_{ij} = \frac{l_i}{l_j} \quad i, j = 1, 2, 3, 4, 5$$

According to the result of the first problem in this task, we can get:

$$b_1 = 0.091 \quad b_2 = 0.482 \quad b_3 = 0.112 \quad b_4 = 0.092 \quad b_5 = 0.12$$

$$l_1 = 0.498 \quad l_2 = 0.165 \quad l_3 = 0.212 \quad l_4 = 0.195 \quad l_5 = 0.091$$

Then the paired comparison matrix are:

$$C_1 = \begin{bmatrix} 1 & 0.1888 & 0.8125 & 0.9891 & 0.7583 \\ 5.2976 & 1 & 4.3036 & 5.2391 & 4.0167 \\ 1.2308 & 0.2324 & 1 & 1.2174 & 0.9333 \\ 1.0110 & 0.1909 & 0.8214 & 1 & 0.7667 \\ 1.3187 & 0.2490 & 1.0714 & 1.3043 & 1 \end{bmatrix}$$

$$C_2 = \begin{bmatrix} 1 & 3.0182 & 2.3491 & 2.5538 & 5.4725 \\ 0.3313 & 1 & 0.7783 & 0.8462 & 1.8132 \\ 0.4257 & 1.2848 & 1 & 1.0872 & 2.3297 \\ 0.3916 & 1.1818 & 0.9198 & 1 & 2.1429 \\ 0.1827 & 0.5515 & 0.4292 & 0.4667 & 1 \end{bmatrix}$$

- (2) Construct the paired comparison matrix of the third layer to the second layer
There are only two factors: the risk and personal issues. Compute the paired comparison matrix by the weight w_1, w_2 , the matrix is:

$$A = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$$

$$\text{Where } w_{ij} = \frac{w_i}{w_j} \quad i, j = 1, 2$$

The value of w_1, w_2 can be determined by the influence of the two factors to the donors' decision. Based on the material we get from the internet, $w_1=0.621$, $w_2=0.379$. Then the matrix is

$$A = \begin{bmatrix} 1 & 1.639 \\ 0.610 & 1 \end{bmatrix}$$

3. Consistency check to the paired comparison matrix

According to the feature of the paired comparison matrix, it is easy to get the conclusion that C_1, C_2 are parity matrix. Through the theorem of parity matrix, the rank of C_1, C_2 are both 1, the single nonzero characteristic root is 5. So any column of the matrix is the characteristic matrix to the characteristic root. Take the last column of C_1, C_2 to be the characteristic vector:

$$W_1 = [0.7583, 4.0167, 0.9333, 0.7667, 1]$$

$$W_2 = [5.4725, 1.8132, 2.3297, 2.1429, 1]$$

$$W_3 = [1.639, 1]$$

Normalize the characteristic vectors above by the following formula:

$$\bar{W}_i = \frac{w_{ij}}{\sum_{j=1}^n w_{ij}} \quad i, j = 1, 2, 3 \dots n$$

So the weight vectors are:

$$\bar{W}_1 = [0.1014, 0.5373, 0.1249, 0.1026, 0.1338]$$

$$\bar{W}_2 = [0.4289, 0.1421, 0.1826, 0.1680, 0.0784]$$

$$\bar{W}_3 = [0.6211, 0.3789]$$

4. Sort of the total layers

Combine the weight vectors $\bar{W}_1, \bar{W}_2$ of the third layer to second layer and the weight vector $\bar{W}_3$ of the second layer to the objective layer.

Construct a matrix $\bar{W}$ with $\bar{W}_1, \bar{W}_2$:

$$\bar{W} = [\bar{W}_1, \bar{W}_2]$$

Then the combined weight vector can be get by the formula:

$$S = \bar{W}_3 * \bar{W}$$

So the combined weight vector of the third layer to objective layer is:

$$S = [0.0630 \ 0.3337 \ 0.0775 \ 0.0637 \ 0.0831 \ 0.6360 \ 0.0539 \ 0.0692 \ 0.1625 \ 0.0297]$$

Through the combined weight vector S , the irrational fears and the probability of survival of the donor are the most important factors that affect the donor's decision. The factors with the least important are level of altruism and level of trust. So if the related department want to motivate people to donation, they can according to the above conclusion and improve the policies which are related to the important factors.

9.3 Section 3

Development to the decision model of donor

When entering a list paired network rather than a direct transplant to the relative of friend, because of the random factor of the paired network, it may affect the potential donor's decision to certain extent. The model is improved based on the first section of task 6.

Import another factor X_7 that affect the decision of the donor. It is the influence affect the decision of donor from the paired network. With the similar method used in section 6.1, get the performance of the paired network and then give value to the factor X_7 according to the investigation and material get from the internet. Divide the probability of successful paired in the current paired network into five levels, and give each level a value from -2 to 2 step by 1. It is showed in the following table:

Table 18: The value of each level of the paired network

The level of the probability of the success in the paired network	Value
$\leq 15\%$	-2
$15\% - 50\%$	-1
$50\% - 70\%$	0
$70\% - 90\%$	1
$\geq 90\%$	2

According to the Logistic regression equation get in 6.1:

$$P'' = \frac{e^{b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + L L + b_n X_n}}{1 + e^{b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + L L + b_n X_n}}$$

And with the same method, we can get the coefficient b_0, b_1, b_2, L, L, b_n of the independent variable.

If the donor know that she/he should donate the organ to the relative through the paired network before donation, the donor can firstly get the probability of successful paired according to the historical data of the paired network. Then the donor can get the value of the factor using table 18. Then use the regression equation and get the value of factor P'' . Accord to the value of P'' , the donor can make decision whether to donate through paired network or not.

Based on the analysis above, we can know that the key to the donor's decision is the probability of successful paired in the network and the coefficient b_7 of X_7 . When the probability of successful paired is too low and b_7 is too small, the donor would be not likely to donate the organ.

9.4 Section 4 Advice to develop and recruit more altruistic donors

Following the development of medical technology, the demand of organ transplant is much more than the supply. The shortage of the organ is the main bottleneck that restrict the development of the technology of organ transplantation. It leads to that too many patients are on the waiting list for a available organ. To improve this situation, more donors are needed to provide more organ. So it is important for the related department to motivate more people to donate. Toward these problems, we give the following suggestion to the related department:

- ◆ Improve the current organ transplantation system, and pay much attention to the donation to relative, and make the relative of the donor have highest priority to get the organ. The donor should be get the prior right and honor from the donation.
- ◆ The hospital that conduct the transplant should improve their technology and service to increase the probability of survival of the donor, and make the donor trust the hospital and avoid the adverse influence as much as possible.
- ◆ Attract more attention of the public, we can recruit young healthy donors through advertisement, public speaking, and word of mouth. Many of our donors have friends or relatives who is the organ recipient and understand how significant their gift can be.
- ◆ Modify the allocation policies concerning the concrete situation of the donor. For example, if the donor pay much attention to the probability of the success of transplant, then it would be better to make the operation of the donor in the place where the probability of the success of transplant is big.
- ◆ The transplant policies in different region of the country can be different. Each region can make the suitable transplant policies according to the transplant situation of the region such as the traffic condition and medical condition.

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11 Appendix

1. Procedure of the synthetical evaluation based on entropy technology
clear all
A=[0.2 0.2 0.8 5;0.3 0.3 0.6 4;0.5 0.5 0.5 3;0.8 0.8 0.4 2;0.9 0.9 0.3 1];
f=[0.2,0.2,0.3,5];
D=[f;A];
for i=1:6

```

        for j=1:4
            B(i,j)=D(i,j)/sqrt(sum(D(:,j).^2));
        end
    end
    B;
    for i=2:6
        for j=1:4
            C(i-1,j)=abs(B(i,j)-B(1,j));
        end
    end
    C;
    a=min(min(C));
    b=max(max(C));
    q=0.5;
    for i=1:5
        for j=1:4
            E(i,j)=(a+q*b)/(C(i,j)+q*b);
        end
    end
    E;
    for i=1:5
        for j=1:4
            Bi(i,j)=A(i,j)/sqrt(sum(A(:,j).^2));

            end
        end
        for i=1:5
            for j=1:4
                Pi(i,j)=Bi(i,j)/(sum(Bi(:,j)));
                Xi(i,j)=Pi(i,j)*log(Pi(i,j));
            end
        end
        Ei(j)=-(1/log(5))*sum(Xi(:,j));
        di(j)=1-Ei(j);
    end
    for j=1:4
        W(j)=di(j)/sum(di);
    end
    ph=[0.2,0.2,0.4,0.2];
    W=(ph.*W)/sum(ph.*W)
    r=E*W'

```

2. Procedure to compute the weight vectors in AHP

```
a=[0.091,0.482,0.112,0.092,0.12]
b=[0.498,0.165,0.212,0.195,0.091]
for i=1:1:5
    for j=1:1:5
        A(i,j)=a(i)/a(j);
        B(i,j)=b(i)/b(j);
    end
end
A
B
w1=A(:,5);
w2=B(:,5);
w3=[1.639,1]
W1=w1/sum(w1)
W2=w2/sum(w2)
W3=w3/sum(w3)
K1=W3(1).*W1
K2=W3(2).*W2
```